

Analysing Icaros:

The Musicology of Amazonian Ayahuasca Healing Ceremonies

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“At this point the old woman begins to sing. Hers is no ordinary song, but an icaro, a magical curing song that in our intoxicated and ecstatic state seems more like a tropical fish or an animated silk scarf of many colours than a vocal performance.”

Terence McKenna, 1992

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[Image I: An Ayahuasca Shaman during a ceremony, holding a Schacapa.](#)



[Image II: The river Amazon going through the rainforest.](#)

Chapter 1

Literature Review

1.1 - Introduction

Indigenous to the Amazon, the term *Icaro* denotes a particular musical performance, exclusive to the shamanic healing tradition known as ‘Ayahuasca’. *Ayahuasca Icaros*, as demonstrable in present-day recordings¹, often come in the form of melodic, verbal sonants, phrases, or motifs, and are often accompanied by other instruments such as rattles and drums (Katz et al., 1971). These musical phrases are employed ritualistically, by a *Shaman*, during the ceremony, which centers around the ingestion of a potent *entheogenic*² tea (Metzner, 1999). The Icaros appear to take on a new form when beheld in the ecstatic state of consciousness, which the plant-brew provokes. Native informants suggest that the icaros play a key role in invoking visionary-type experiences, which in turn, influence other shamanic tasks such as diagnosing psychosomatic illness and attempting to cure it, or alleviate any symptoms³ (Luna, 1984; Katz et al., 1971). The intention of this research is to acquire a collection of audio recordings from ayahuasca ceremonies, and conduct an in-depth analysis of its musical components in order to further understand the nature of a type of music, employed as a therapeutic medicine in its indigenous context.

Chapter One explores the current literature surrounding Icaro use within its ritual context.

Chapter Two explores the methodologies of previous, similar research, and the methodology of this research.

Chapter Three exhibits all the analysis of the Icaros, ready to be discussed in **Chapter Four**.

Chapter Five will summarise and seek further discussion.

¹ The available recordings can be found in section 2.3.

² *Entheogenic* is a modern term for ‘hallucinogenic’ or psychedelic which attempts to include a sense of spirituality within its definition. The etymology suggests a meaning of ‘generating the divine within’.

³ Special weight is placed on the success of Ayahuasca in dislodging previously over-trodden thought patterns which cause addiction or depression-like issues. (Labate & Cavnar, 2013; Mabit, 2007)

1.2 - Icaros

On a rudimentary level, an Icaro is a song that is performed within the, now globally known (Tupper, 2006), Ayahuasca ceremony of Amazonian shamanism. The term appears to have a unique cultural origin, as the word itself appears across many different cultures that surround the operation of Ayahuasca healing ceremonies (Brabec de Mori, 2011), and the greater Amazonian Basin in general. The word '*Icaro*' is suggested to be linked to '*icarar*'; the *Quencha*⁴ verb meaning 'to sing' (Bustos, 2004. Caldicott (N/A) further suggests that there may be an association with the *Quencha* verb '*curar*', meaning 'to cure'.

These melodies that strictly accompany the Ayahuasca Ceremony, are sung, chanted, whistled or hummed; It is also known to be played on ethnic flutes, and even violins (Uzendoski, Calapucha-Tapuy, 2011). These Icaros, while analogous to music, are performed and experienced in a way that is incomparable to western musics, and indeed many other musics from the globe. The primary purpose of performing Icaros (in the belief of those who perform and experience them) is to perform spiritual tasks. While this too, is arguably an artistic expression through sound (therefore seemingly analogous to music), it is important to consider that the philosophical motivation behind the performance of Icaros does not necessarily concern any sense of aestheticism, emotional expression, or even the concept of a musical product (a finished song). Instead, the Shamanistic intentions at the heart of the performance suggest that if there is any 'product' being sought, in the same way a composer may envision their masterpiece, it is a Shamanic product (for example the curing of anxiety) Therefore Icaros can be argued as an expression of shamanism, not music, and therefore cannot be compared to other musics, as the application of air pressure changes is not being performed for the same reasons at all. However, the use of sound can still be studied and examined under the lense of musicology, as examining these Icaros in this way has been effective previously.

Previous research on the Icaros of Ayahuasca ceremonies, predominantly feature the work of Musicologists Fred Katz & Dobkin de Rios, undertaken in Peru (Katz et al., 1971). By collecting data and analysing the Icaros they had recorded '*in-situ*', they extended the idea of a 'spiritual ladder', conceptualising a model dubbed 'the Jungle Gym in consciousness' (Katz et al., 1975). This theory suggests that the subjective manifestation of Ayahuasca Icaros, in an ecstatic state of consciousness, is that of a "jungle gym" style structure. To further define this, the authors suggest that the shamanic performance creates an inflexible, secure structure, which allows the user some freedom to engage in independent pathfinding and exploration. At the time of the research by Katz and de Rios, access to the music was scarce, with limited ability to 'upload' and share recordings or transcriptions of the music. Thanks to the advances in

⁴ Quencha refers to the 'singing mountaineers' of Peru. (2010, Arguedas)

technology, there is quite an abundance of recordings available online where fieldworkers and researchers have visited the location and captured the recordings live.

Luna (2006) places a lot of stress on the idea that the very melody of the icaro itself contains the healing ‘power’, which explains why Caldicott ponders upon the connection. The respective people believe that it is the ‘spirits of the plants’ that teach the Ayahuasca Shamans these melodies. The most adept Ayahuasca Shamans will have an extensive archive of Icaros, or healing melodies, memorised, from which they pick specific ones, dependent on the intention. This can include such things as attempting to rectify physiological or emotional side-effects of ayahuasca, or provoking transitions between different ‘phases’ of the subjective experience. On a more speculative level, these songs are used to commune with the ‘plant-spirits’ linked to the indigenous folklore cosmology of ‘supernatural agents’ (Gow, P., 1996).

Susanna Bustos studied Ayahuasca Icaros from the perspective of phenomenology, and states that, according to her informants, the experience with Icaros stand out as a pivotal moment within the healing process that unfolds during a ceremony (2008). As the most revered and profound technique for aiding the participants in their asomatous travels, Icaros, acting as an ancillary mechanism within this folklore healing tradition, draw the main attention of this research. Deciphering the musicology of Ayahuasca may lead to understanding more about what defines a ‘healing song’, and perhaps in later studies, which musical components bestow a song with ‘healing qualities’.

1.3 - Ayahuasca

Ayahuasca is the *Quencha* name for the ancient healing ceremony that has existed amongst indigenous people for millennia in the Amazon (Luna, 1991; Tupper, 2006; Vickers, 1993). Additionally, ‘Ayahuasca’ is the cultural name of the jungle liana *Banisteriopsis Caapi*; ‘The Vine of the Souls’ in native belief. From here on, the vine *Banisteriopsis Caapi* shall be named B. Caapi to avoid overusing the word Ayahuasca, and overcomplicating the word’s meaning. The ceremony itself focuses on the digestion of an entheogenic tea made from B. Caapi. The shamans believe that, with correct diet, ritual and state of mind (ideally the state induced by the plant-brew), it is possible to achieve communication with plant-spirits for the purpose of diagnosing, curing afflictions, treating psychosomatic health issues, purging the physical body, and other shamanic tasks such as prophecy-divination (Wagley, 1974). In fact, communion with the spirits of plants is a central theme in Amazonian cosmology, with informants claiming that the original ayahuasca brew was taught to the natives by ‘plant teachers’ (Luna, 1984).

Bustos (2004) divides the ceremony into four phases which include two discussion phases (one at the start, one at the end), where the participants and the residentary shaman speak. The two phases between these discussions include a 'non-verbal working phase', in which the Shaman 'charges' the previously prepared potion with exemplary curing songs and tobacco-smoke blowing, followed by the 'main phase', in which the concoction is consumed and a heavy use of Icaros occurs. The same account informs that the ceremony is conducted in the dark, commencing as the day approaches twilight. As the effects of the concoction can remain active for over 6 hours, the ceremony can last extensive periods of time.

During his time observing four informant mestizo ayahuasca shamans in Iquitos, Peru, Luna (1984) ascertained a sense of the local cosmology. He describes their belief in a bilateral cosmovision consisting of two parallel worlds of earth and water. He further adds that there is acknowledgment of a third, higher plane, but this contributes far less to their mythology and legend. The more relevant mythology and cosmology includes the perception of 'plant spirits', best worded by Pablo Amaringo⁵ in the foreword to H.G Charing and Peter Cloudsley's book 'Ayahuasca Visions' (2011).

"Every tree, every plant, has a spirit. People may say that the plant has no mind. I tell them that the plant is alive & conscious. A plant may not talk, but there is a spirit in it that is conscious, that sees everything, which is the soul of the plant, its essence, what makes it alive. The channels through which the water & sap move are the veins of the spirit."

1.4 - The Brew

The archaic brew is prepared by boiling together multiple indigenous plants in a pot for several hours. It is often referred to as 'the purge' (or '*purga*') to natives, due to the powerful purgative quality. (Mabit, 2007) Dobkin de Rios theorised that this 'nausea', and other 'bad trip' qualities are caused by rapid access to the unconscious. This characteristic is associated with the bodily cleansing that ayahuasca is revered to cause, and contributes to the duality described by many personal accounts, in which there is suggested to be a dichotomy between feeling liberated from the body which is so susceptible to nausea, vomiting; a humbling experience. (DMT Nexus, 2015)

⁵ Pablo Amaringo is an acclaimed artist who explores the concept of Ayahuasca intoxication in the main body of his work (Luna et al., 1991). He has worked with researchers, as well as having written his own book. He attempts to depict his visions whilst under the influence of the brew.

The vine *Banisteriopsis Caapi*, is the fundamental ingredient in the preparation of *Yagé* in a large majority of Ayahuasca ceremonies. It is mixed with other plants such as *Psychotria Viridis* which contain N-Ndimethyltryptamine (DMT)⁶, and other hallucinogenic substances (Strassman, 1994). *B. Caapi* contains *harmaline*; referred to in the field of pharmacology as an inhibitor of monoamine oxidase⁷ (Demange, 2002). These MAO inhibitors prevent the deamination of DMT, and this is the particular chemical mechanism which determines the consistency for which various banisteriopsis species are used in conjunction with other plants. (McKenna & Towers, 1984) Deliberate restraint has been used to discuss the chemistry, as the understanding of this is not crucial to the research here, the summary which follows this line elucidates the purpose of it being raised. For a more accurate and detailed understanding of the chemistry please refer to the work of Strassman (1994).

Luna is one author to list the many alternatives to *P. Viridis* including: Tobacco, Tangarana (*Triplaris Surinamensis Chamiso*), Ayahuma (*Couroupita Guianensis Abl.*) and many others. During the creation of the concoction, the shaman must sing certain icaros, blow tobacco smoke and perform the equivalent of reciting prayers or blessings.

In his first chapter of his own book, Amaringo delves into great depth and detail about the long, spiritual process of prepared the brew, including using intuition and spiritual communication with '*chicua*' birds for just selecting which plant to use; constantly avoiding interaction with bad spirits or malevolent forces (Luna et al., 1991). Once the plant has been sighted, the shaman will give offerings of tobacco, banana leaves and matches to the grandfather spirit, asking permission to take the ayahuasca vine for medicinal purposes. Many authors describe a necessity to respect the depth of which the natives believe ayahuasca to be spiritual and sacred. During ayahuasca visions, any questions are directed with great respect to the spirit of ayahuasca (the grandfather spirit), such as 'please help me to see that which I require to see'. Amaringo also describes a strictness about diet, also uniformly noted by other researchers. The diet involves avoiding salty, sweet or bitter foods, no drinking or drugs, no speaking or interacting with bad-living people. This diet lasts a few days prior to the preparation of the ceremony, and it is somewhat reflective of the extensive diet which the shaman follows for many years during his training to be an ayahuasquero. Much of the ideology of the native amazonians, including the commonly associated imagery, symbolism and mythological meaning is presented in Amaringo's vivid paintings.

⁶ *Dimethyltryptamine*, among other *tryptamines*, is a potent hallucinogenic substance. As mentioned before in the text, it is perhaps one of the most potent known, even more so when combined with the appropriate enzyme inhibitor. Rick Strassman (1994) appears to be the authoritative figure on the chemistry and neuroscience of hallucinogens, especially DMT.

⁷ As a side note it is worth mentioning that although the unique MAO inhibiting property is found in the *B. Caapi* vine, the harmaline itself is also psychoactive in the human body.



[Image III: Brewing Ayahuasca](#)

1.5 - The role of the Shaman

Each ceremony is conducted by a shaman, often referred to as an *ayahuasquero* or *vegetalista*.⁸ The main role of this Shaman is that of an overseer, spiritual guide and emotional counsellor. It is their primary responsibility to create the magical brew, and ensure the physical, emotional and spiritual safety of the participants during its consumption (Bustos, 2008; Metzner, 1999; Katz et al., 1971). A Shaman's ability to achieve these three tasks, can be considered to represent their 'power' or 'ability', and Bustos states that the main expression of an Ayahuasca *Curanderos*⁹ power to facilitate the ceremony, is the use of Icaros (Bustos, 2004).

⁸ Luna notes that '*vegetalistas*' is more accurately understood as an umbrella term for various plant-medicine practitioners such as *Tabaqueros* (Tobacco based practitioner) '*Purqueros*' or '*Ayahuasqueros*' (Ayahuasca/Icaro based practitioner) and others. (Luna, 1986)

⁹ Susanna Bustos' (2008) preferred term for Ayahuasca Practitioner. Curandero means 'folk medicine practitioner'.

Why music (in the form of Icaros) is a powerful *tool* in the context of Ayahuasca can be explained with the discussion of Altered states of consciousness, particularly the studies of the ‘ecstatic’ state of consciousness achieved by the participants during an Ayahuasca ceremony.

1.6 - Shamanism & Ecstasy

The study of augmented consciousness is of prime relevance in this exploration of ayahuasca, as the induced state of *ecstasy* is central to the source of the healing. As the title quotation¹⁰ suggests, in the ‘ecstatic state’ caused by the DMT-fueled experience, the music can manifest in bizarre forms to the user’s subjective perception (McKenna, 1992; Leary, 1964; Strassman, 1994). A state of ecstasy (or an altered state of consciousness), can be described as a state of contemplation and awareness, coupled with sensory-overstimulation and hallucinations.

Gilbert Rouget (1985) extensively attempts to contain the characteristics ecstasy in one single definition, but ultimately concludes that only severe reductionism can bring about an agreement across all language and history.¹¹ Based on Rouget’s etymology, the word ‘ecstasy’ can be traced to the word ‘ἔκστασις’ (ek-stasis), which holds a meaning similar to: ‘to stand (*static*), outside (*ex-*)’. This, coupled with other authors definitions¹², suggest that ‘ecstasy’ refers to ‘flight of the soul, from the body’, ‘exist- outside- the mind’ (Hanegraaff, 2012; Devereux, 2008; Friesen, 2011). Aldous Huxley (1954) coined the term ‘psychedelic’, a word which attempts to describe the mind manifesting before the soul.

Mircea Eliade (1964) observes similarities in global Shamanistic culture and suggests that regardless of whether it be the use of the hallucinogenic *Peyote Cactus* in North Mexico (Leary, T., 1964; Castaneda, C., 1968); the consumption of potent *Amanita Muscaria* mushrooms in Siberian and early-Vedic Shamanism (Furst, P., 1976; Devereux, P., 2008); *Ibogaine* and *Datura* ingestion in African spirit mediumship (Beattie, J., Middleton, J., 2004); or the drinking of ayahuasca tea in the Amazon; the aims of the Shaman in consuming and interacting with hallucinogenic plants, is to master the techniques of inducing, controlling, nurturing and exploring ‘ecstasy’. McKenna (1992) propounds that even the non-plant consuming shamans across the world are attempting to give rise to ecstasy through other means

¹⁰ “At this point the old woman begins to sing. Hers is no ordinary song, but an icaro, a magical curing song that in our intoxicated and ecstatic state seems more like a tropical fish or an animated silk scarf of many colours than a vocal performance.” (McKenna, T., 1992)

¹¹ Considering this, ecstasy and trance will be used interchangeably, to describe an alteration in consciousness without attempting to confirm which state, specifically, which the mind ends up in during

¹² Friesen (2011), Rouget (1985) and others at length discuss the theoretical definitions of difference states of consciousness.

such as drumming or fasting. This extends Eliade's deduction, further propounding that *all* practice of shamanism worldwide, is the inquest upon the state of ecstasy.¹³

Ayahuasca Shamans, in light of Eliade's theories, are masters of the ecstatic state, and express their power over ecstasy through the chanting and singing of Icaros.

1.8 - Music & Altered States

The use of Icaros as a tool to assist and facilitate a state of ecstasy, is an example of how music can be directly used to manipulate the brain. This accurately defines the passive motivation for examining Ayahuasca Icaros. As an isolated example of music directly manipulating the brain, there is potential that examining the music used may reveal more information, or yield a new understanding of consciousness, based on how consciousness reacts and changes when subjected to music.

Gabonian Iboga initiation ceremonies are an example of where music is used to manipulate or facilitate subjective effects of hallucinogenic substances. In these ceremonies, participants digest iboga root¹⁴ and are subjected to polyrhythmic drumming which induces a powerful trance or 'spirit possession' (Maas, et al., 2009). The researchers attempt to decode the polyrhythmic drumming in order to find out which rhythmic conditions had to be met in order to entrance a Mitsogho¹⁵ participant. Researchers who have observed this practice of spirit mediumship have been informed that music acts as a safety-rope between this world and 'the hereafter', and serving as a form of locomotion between the two realms (Maas, et al., 2009; Beattie, J., Middleton, J., 2004). Jeremy Narby (1999) further extends the notion of symbolic repetition when music features as a significant component of a hallucinogenic ritual, in which he cites Maria Eliade's compilation of "countless examples" of various forms of 'shamanic ladders' or 'spiritual ropes'¹⁶ (Eliade, 1964).

Ethnomusicological data gathered regarding the Gabonian Iboga initiation and in particular the data concerning rhythm has been of use to other researchers who study the synapse between perception and consciousness, such as Csaba Szabo (2009), who details the effects of monotonous drumming on 118 students in England. While the data here wasn't attained by

¹³ "The rational, mechanistic, anti-spiritual bias of our own culture has made it impossible for us to appreciate the mindset of the shaman. We are culturally and linguistically blind to the world of forces and interconnections clearly visible to those who have retained the Archaic relationship to nature." McKenna on understanding the role of the Shaman (1992).

¹⁴ A hallucinogen.

¹⁵ Name of the tribe: The Mitsogho tribe of Gabon.

¹⁶ The term 'Spiritual Ladder' refers to the previous notion of a safety-rope between 'two co-existing realms'.

fieldwork, it is clear that there was a considerable effect on consciousness when the music was played, and similar studies in the future can be done to compare the effects of Gabonian trance-inducing polyrhythms, to the effects of the monotonous drumming used previously in this experiment.

In the case of the Adyghs¹⁷ (*'Circassians'*), music acts as a medicine through alleviating disharmony between person and nature (Sokolova, 2007). Sokolova also indicates that optimistic ceremonial music acts as anesthesia, citing the story of a War Hero, whom had a bullet removed from the leg in this manner. Many archaic and traditional cultures believe disequilibrium between person and nature is expressed as illness or disease.

It is not uncommon for cultures to centre psychotherapy or medicine around the use of drugs and/or music, even witnessed in waves of Western culture. In a study of the House/Ecstasy movement, Metcalfe (1997) discusses how the movement, based on psychedelic drugs and music lead to the transcension of social codes such as gender, sexuality and race on the dancefloor, in a way that had yet to be experienced before.

¹⁷ An Iberian-Caucasian group found in North Caucasus.



[Image IV: Preparing Ayahuasca Brew](#)



[Image V: Ready to brew](#)

Chapter 2

Methodology

2.1 - Motivation and Intention

The motivation for this research, if not ascertained from the literature review, is to contribute to the greater understanding of how music affects the brain and consciousness. By breaking down shamanic curing songs into such constituent parts as melody, rhythm, harmony, instrumentation, performance dynamics and structure. There is a potential for patterns to form, or constants to be observed, which define the musical style. As this music is associated with therapeutic healing, understanding the constituent parts may reveal more about how any ‘musical component’ interact with human consciousness.

Previous examples of Ethnomusicology essays (Maas, et al., 2009; Dobkin de Rios; 2009; Sokolova, 2009), demonstrate how obtaining empirical results in this way can develop into theses when combined with the developing theories of other researchers. Szabó (et al., 2009) conducted an experiment where consciousness-affecting polyrhythms were used to measure the phenomenological impact on perception¹⁸. In a similar way, recreating the Ayahuasca paradigm within a controlled setting, using recreated Icaros (based on the findings on this dissertation) may be useful to explore the psychological and chemical mechanics involved with Ayahuasca-intoxicated perception. Due to its close proximity to other fields of study, the line of research regarding perception of sound and its contribution to consciousness, has the potential to include further musicology, psychology, biology and neurochemistry¹⁹.

¹⁸ 118 University Students were subjected to these recordings (210 bpm polyrhythmic drumming, perhaps influenced by empirical data taken from ethnomusicology studies)

¹⁹Including the study of how chemical reactions in the brain [such as those produced by perception of sound and the associated psychological mechanisms] manifest as changes in consciousness.

Should studying Ayahuasca curing music, which manifests as such a profound experience in the reports of its beholders, reveal more about how perception of sound affects consciousness, then there is reason for further research into, not just Ayahuasca, but Global Shamanic Music in its entirety.

2.2 - Transcription as a tool

Similar ethnomusicological research on Ayahuasca Icaros has been undertaken in the past by the researchers mentioned in section [1.2](#) (Katz et al., 1971; 1975; Bustos, 2004; 2008) and before approaching the data acquisition stage, it would be useful to consider the methodologies employed previously. Dobkin de Rios used transcription to describe the unconventional performance style of the whistling; primarily following western notation with a few additional markings. Within this score she denotes rhythm, pitch, pauses, phrasing and also attempts to describe the structural form. While the ability to record is cheaper now, and recording technology offers a simple solution to capturing live performances and saving the information for later, there is still much value in transcribing the music in this way.

Bruno Nettl (2005a), a reputable musicologist, discusses at length many of the issues that the working ethnomusicologist faces. In Essay 7, Nettl discusses how transcription acted as the main tool of the ethnomusicologist for many years, and even as recording technology arises, is still heavily relied upon today. It is suggestible that a desire to translate non-western music into western notation, is an attempt to translate their music into a language that can be understood in enough sophistication and detail [by the observer] to truly acknowledge the complex musical ideas. In the same way western-trained musicians become very familiar with the subtle nuances of classical harmony, violin timbres and the English language (as a few examples), it has to be expected that western (but not strictly limited to western) fieldworkers can never ascertain every detailed nuance pertaining to that musical expression of culture, simply because they aren't an integrated member of the community with as much exposure.

Within this Ayahuasca research, it is clear that the Icaros are non-tangential to a western understanding of music, therefore utilizing the western score as a method to attempt to understand the music, does not attempt to suggest anything more than a western understanding of non-western musics. Should the scores not accurately relay the music, if ever recreated from the scores, it would leave speculation that the 'functional healing properties' of the music is something far less tangible to the western mind than playing certain notes at the necessary time. This in itself would be a strong development into understanding how 'healing music' works in essence.

When considering previous methodologies, rather, any ethnographic work, there must be special consideration of the human aspect. As suggested in the notions of symbolic and interpretive anthropology (Geertz, 1973), studying any kind of culture must be regarded as interpretive, rather than experimental. Recognising that each ethnographer, researcher or fieldworker are themselves a form of actor portraying a perspective, no single author can be relied upon to relay the 'true' happenings of indigenous cultures. With this issue in mind, the transcription seems ideal for a translation-style role, as the issues of not being able to truly relay the non-western music is intrinsic and unavoidable in ethnomusicology. Following the logic of Nettl and Geertz, it is clear that one's ability to transcribe is an area in which interpretation can massively affect how this music is perceived by readers, as a fundamental problem in academic musicology stems from an archaic dependence on 'proving the music exists in this way' by attempting to write it down. (Nettl, 2005a) Non-written music, such as folk traditions and word-of-mouth music cultures, is automatically skewed with any attempt to write it down, yet it appears to be a destructive, but necessary, process for the music to be recognised by western scholars.

2.3 - Problems of the Ayahuasca Ethnomusicologist

Nettl (2005b) writes along similar lines in another essay, suggesting that, due to their position, fieldworkers, such as those who have gathered the recordings needed for this research, affect the field so much that they can get in the way of their own research. The very presence of the researcher themselves has an influence of the culture being observed. Operating remotely should avoid further disruption of the field, sidestepping this issue of affecting the field, just by being there, however, there are further issues which is relevant to the data acquired for this research.

The location and characteristics of the ceremony makes high quality, clear recording of the music difficult. There is much movement around the participant group, causing difficulties in fluent and clear recording of the melodies. Also, many Ayahuasca shamans disapprove of being recorded, and even when permission is accepted, the ceremony lasts for extensive periods of time making it difficult to take palatable recordings that accurately define the varying musical performances throughout. Furthermore, the location in the rainforest also directly contrasts ideal recording conditions, largely because of the immense background noise present in every recording, when combined with leafy rattles there is little spectral space for some of the lower frequency chanting to remain clear. However, as the task is to analyse what the participants of ayahuasca ceremonies are hearing, the dominant presence of rainforest background noise cannot be discounted as an influential sound source, so it could be regarded as valuable information for realising the full sound content that is experienced during an Ayahuasca ceremony. This constant

influx of aural information must contribute to the state of the set and setting²⁰, and should be included when analysing audio content in all of its components.

Dynamic information is also lost to many recordings, therefore it is better to avoid speculating on dynamics and their relevance as the data is not reliable (is the shaman singing quieter; or has the background noise gotten louder; or has the shaman moved away etc). In the data acquired, there are examples of instances where rhythmic or harmonic content is dramatically reduced or inaudible simply because assistants and performers may be stood too far from the limited microphones. However, enough high quality data has been taken which accurately portrays the chosen categories.

2.4 - Analysis Categories

As suggested in section [2.1](#), the methodology for decoding Ayahuasca Icaros shall include transcribing the available recordings into complex extended scores, aiming to be transparent and avoid losing musical ideas through the process of translation. Utilising a multitude of extra markings, the scores will capture enough of the musical information to be interpreted by other musicologists, without overlooking musical ideas or musical philosophies exclusive to this music style. In a sense, recording technology helped to avoid this (Downey, 2002), capturing ‘exactly’ what is being played, which allows objective listeners to attempt to decode the musicology without the pressure of being an attending presence that impacts the proceeding events (Nettl, 2005a).

Thus, the upcoming specific categories for analysis will be of high priority, but do not mark the limit to what musical information is believed to be relevant. Information such as performance dynamics, may be a category that is only relevant when there is significant use or application of.

2.4.1 *Melodic Content*

Melodic content refers to pitch and melody, and describing the spectral narrative and physical sound qualities of the recordings. This can include favoured or recurrent modalities or scales, as well as voicing characteristics²¹. Dobkin de Rios (2007) states that *frequency* appears

²⁰ Set & Setting are key components of any hallucinogenic experience. According to Leary, the set describes the mental, emotional state of the person, and the setting describes the surrounding environment. The conditions of these two factors most accurately defines the nature of a trip, such that, if the mental state is bad (angry, stressed, scared), and the environment is bad (exposed, controlled, uncomfortable people present), the experience is likely to manifest as bad (Leary, et al., 1964).

²¹ Such as vocal range, voice type (where in the spectrum the voice sits)

to be the lowest common denominator of Icaros, suggesting that, the frequency of rattling or rapid vibratory sounds in combination with whistling or singing or chanting are very important to the healing experience. To elucidate this statement, the author suggests that, within the context of her previously introduced *Jungle Gym in Consciousness* model, the frequencies produced here act as the modular building blocks which assemble into the fixed-path, enclosed construct.

The method for this will, as suggested in , involve transcribing the recordings, notating in a style that attempts to remain as conventional [as a synonym of simple, affordable] as possible, but making use of alternative notation techniques typical to ethnomusicology. The characteristics of the music are described as different and isolated from other local, popular music styles, which are themselves in high contrast to western style²². Therefore the expectation is that there will be a significant dependence on these alternate markings to identify potential patterns. By, for example, drawing the curvature of the melody, there may prove to be geometric motifs that identify how a patterns develop.

2.4.2 Rhythmic Content

Maas and Strubelt (2009) undertook research into the role of Rhythmic content in music that creates or accompanies hallucinogenic experience or trance. Their research offers many useful insights into the potential roles of rhythm in ayahuasca, as well as insights into potential methodologies best used for examining Ayahuasca rhythms. Their paper entitled ‘Polyrhythms Supporting a Pharmacotherapy’ observes a ritual that occurs in Gabon, Africa, referred to as the ‘Iboga Initiation Ceremony’. In this scenario, the participants consume pieces of ‘iboga root’²³ bark until it induces a powerful, ecstatic (out-of-body) experience, or ‘near death experience’, for the sake of healing or spiritual progression. According to various accounts of Gabonian Mitsogho²⁴ informants, the music that accompanies this ceremony is of great importance, acting as an interdimensional ‘safety rope’ that serves as a form of spiritual locomotion. Some of the analysis that is offered by the authors suggests that, during the trance inducing phase of the ceremony, there is an increase in rhythmic changes, which stimulates the cerebellum and hippocampus in the same way as the Ibogaine substance, and so is assumed to augment the effect of the Ibogaine.

The technique employed to trace drum patterns included using grids to identify the individual rhythms and meters that overlap within an overall 12-beat pattern. As Nettl (2005a)

²² Katz (et al., 1971) compares Icaros to Gregorian Chants, in that, they do not feature outside of their ritual/religious context.

²³ ‘Ibogaine’ is the psychoactive compound used heavily in various african cultures, found primarily in the bark from the roots of the Iboga plant.

²⁴ The tribe whom have been documented to engage in this use of Iboga root and polyrhythmic drumming

identified, the Western mind can struggle to deal with Rhythm with as much sophistication as harmony and melody, due to the nature of Western music and its focus on melody.

Another example is found in a study by Csaba Szabó (2007), in which an experiment is conducted to determine the effects of monotonous drumming on subjective experience. In a dimly-lit laboratory environment, 118 university students were individually subjected to 30 minutes of monotonous rhythmic drumming and told to close their eyes, and to imagine a hole leading down into the earth, (an example of Chichosz's (2014) attentive concentration²⁵). The author concluded that exposure to the 210 pulse per minute drumming had a substantial effect in altering perception of time, perception of changes in body-image and the image of the world around, and with it, the meaning of such things. Some participants had strong experiences of reliving moments from the past, and having sudden emotional outbursts such as crying due to such experiences.

Within the framework of my own research it will be hard to speculate on neurophysiological concepts in the same way as Maas & Strubelt, as there is no access to an Electroencephalogram (EEG), or any other means (besides phenomenological approaches²⁶) to measure brain activity within the context of Ayahuasca ritual. But, due to the importance of rhythm in this and other shamanic ritual, it will be valuable to assess the rhythmic content of the Icaro singing, chanting and percussion as this specific content has already been discovered to be crucial.

2.4.3 Harmonic Content

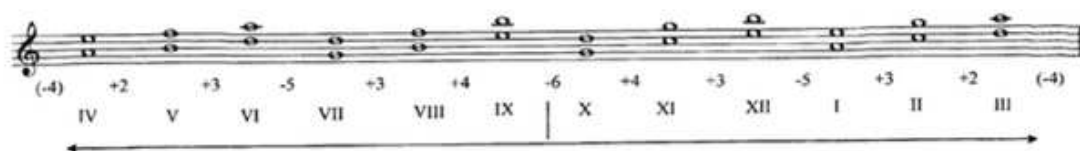
Harmony, as noted by Scherzinger (2013), has been overlooked in the past, due to a preoccupation with rhythmic content, particularly regarding the kinetic nature of it. In Scherzinger's paper 'Fractal Harmonies of Southern Africa' (2013), *mbira* and *matepe*²⁷ music, stated to be linked with spirit-possession, is explored from the perspective of harmony. The method used by Scherzinger to examine harmony, mainly revolves around transcribing the performance into a western-like score, which aids the researcher in identifying a pattern of interest. In short, the *mbira* and *matepe* player's individual notes formed a dyadic sequence that moved in intervals. Perhaps only identified by the process of transcription, Scherzinger observes an '*intervallic palindrome*', whereby the intervals moved form a reversible sequence.

²⁵ See section [5.2](#)

²⁶ Such as that engaged by Bustos (2008).

²⁷ These instruments have been known to be associated with conjuring ancestral spirits.

Figure 5. The Intervallic Palindrome in the Dyad Sequence of *Aroyiwa Mwana*



Example of 'Intervallic Palindrome' described by Scherzinger (2013)

Another justification for including harmony stems from the concept of *binaural beats* and *beating*. Already renowned for its implementation into psychotherapy, the subtle amplitude modulation-like effect of layering two audio signals of a similar frequency together, known as *binaural beating*, is an example of how harmonic information contributes to brain functionality (Lane, et al, 1998). Should there be an extensive use of very dissonant harmony, it could indicate the deliberate application of mind-altering binaural interplay.

As the melody will be depicted by notation, and make use of ghost curvature in describing the melodic narrative in more depth, harmony could be incorporated into the score with ease as a second instance of drawn curvature. Roughly describing the harmony present in the recordings using this method will reveal how often it is used; how intermittently or persistently the occurrences are. Overlapping of the melody & harmony curves will show complete or dissonant harmonies which can be commented upon at a later stage.

2.4.4 Instrumentation

As instrumentation is more identifiable, making a list of the recurring performance instruments, taking into consideration the family of which they each come from, should clearly identify the significant timbre or instrument family. Once the significant instrumentation family has been identified, further review of literature may reveal an insight into the reason for choice.

It may prove that the instrument playing the melody is unimportant, and that it is indeed the musical content itself, and something regarding the perception of that music that triggers its healing effects. Luna (1984), as mentioned multiple times within this text, states clearly that the natives believe that the melody itself holds the healing properties, and this would coincide with any assumptions that specific instrumentation is less relevant in the overall proceedings.

However, due to the scarce nature of any papers questioning the relevance of instrument/timbre types in trance-related or otherwise shamanic music, it is currently difficult to speculate, and it could be an entire new dimension of music perception to explore, comparing the

differences between trances/possessions induced by non-pitched drums and pitched percussion, chanting icaros and incantations and so on.

2.5 - Recordings Acquired

The following repertoire has been acquired for the purpose of analysis and decomposition. As the proper accreditation is due, the recordings will be organised by Publisher.

2.5.1 - H.G Charing & P. Cloudsley

Howard Charing has spent the last 24 years becoming an acclaimed researcher in the field of shamanism, with extensive time spent in Peru with ayahuasqueros. Charing has co-authored multiple books regarding various aspects of amazonian culture, in particular the arts surrounding ayahuasca (Heaven, et al., 2006). Peter Cloudsley, co-author of ‘The Ayahuasca Visions of Pablo Amaringo’ (2011), and is known internationally as a musicologist, whose recordings have been featured by the BBC and on CDs.

In their time they have acquired some of the highest quality recordings available, which focus on the vocal performances of each informant shaman. Secondary information including harmony parts and percussive parts are not always as clear as the main voice, yet have been captured in enough detail to speculate about the role it plays (although at times it is distant and only vaguely audible). There are a total of 8 recordings of Ayahuasca Icaro performances, each varying between 4 minutes and 10 minutes and include some useful information such as the particular sub-amazonian culture the performer adheres to (Mostly Shipibo and Huitoto²⁸). These recordings are published on the personal ‘SoundCloud’²⁹ of H.G. Charing, and I have personally contacted the owners of the recordings and have acquired permission to transcribe and examine each article, and have received confirmation that they are indeed ‘in-situ’ recordings (Charing, 2015).

2.5.2 - SpiritQuest Center

Located on the banks of the Río Momón, an hour from Iquitos (Peru), the *SpiritQuest* center offers a retreat for local and international ayahuasca seekers, which work closely with the

²⁸ Huitoto (Also: *Witoto*) and Shipibo people are indigenous tribes of the Amazon rainforest. Both are known to engage in Ayahuasca ceremonies.

²⁹ A popular music hosting and sharing website. www.soundcloud.com

native communities' shamanic traditionalists to conduct secure and peaceful ayahuasca ceremonies in their intended location. The sanctuary acts as a hub for many different holistic, mystic or natural healing methods and provide much information for those who are interested. Due to the close relationships developed with the shamans, they have been able to capture and acquire many 'in-situ' recordings.

In order to differentiate the source of the recordings, I will also be utilising the library of recordings available from the *SpiritQuest* Ayahuasca centre. It is likely that, while mainly focusing on the extremely high quality, and more recent recordings available from Charing and Cloudsley, using the vast array of shorter recordings to test consistency of any observations. Such that, if a particular scale is identified as a constant amongst the HG Charing recordings, it will be of great use to then consider the *SpiritQuest* recordings comparatively.



[Image VI: Fanning the flames](#)

Chapter 3

Analysis

This Chapter contains the analysis of several Ayahuasca Icaros, based on the acoustic analysis of the recordings acquired (see [2.6](#)). This section will contain scores and graphs, as well as text that attempts to fully describe the musical content of each recording. Sections [3.1-3.4](#) are Icaros taken from *H.G. Charing*'s Soundcloud and have been analysed using the transcription style, as described in section [2.3](#). Sections [3.5-3.7](#) are further examples of Ayahuasca Icaros, in which recordings available on the *SpiritQuest* website are examined more holistically.

Please note that uppercase-lettering marked in *italics* refers to the name of the phrase being discussed, and any uppercase lettering marked in **bold** refers to musical pitches (frequencies).

H. G. Charing Recordings

3.1 - Ayahuasca Icaro Benjamin Ochavano

Published by H.G Charing on Soundcloud: www.soundcloud.com/hgcharing

Track Length: 10:08

Benjamin Ochavano Ayahuasca Icaro

published on SoundCloud
by H.G. Charing

A b b

B {Long Pause}

A²

B³

C 1:00

A³

B³

C² 1:40

A⁴

ARCHIVES

The image shows a handwritten musical score on a spiral-bound notebook. The title 'Benjamin Ochavano Ayahuasca Icaro' is written at the top. Below it, a note says 'published on SoundCloud by H.G. Charing'. The score consists of ten staves, each with a letter and a superscript (A, B, A², B³, C, A³, B³, C², A⁴). The first staff (A) has a key signature of one flat (B-flat) and a common time signature (C). The music is written in a simple, handwritten style with notes, stems, and beams. There are yellow and green highlights under some of the notes. A bracket labeled '{Long Pause}' is written next to the B staff. Time markers '1:00' and '1:40' are written next to the C and C² staves respectively. The word 'ARCHIVES' is written at the bottom left of the page.

Handwritten musical score on 12 staves. The notation includes eighth and sixteenth notes, rests, and bar lines. The staves are labeled with letters and superscripts on the right side, indicating a sequence of chords or sections: B⁴, A⁵, B⁵, C^{*}, D, A⁶, B⁶, C³, A⁷, B⁷, and C⁴. The sixth staff begins with a handwritten instruction "(Long Pause)" followed by a treble clef and two sharps (F# and C#). The notation is written in black ink on white paper. The bottom left corner of the page features the logo "ARCHIVES" and the text "125-12 Stave".

Section II

A

B

C

B²

C²

B³

C³

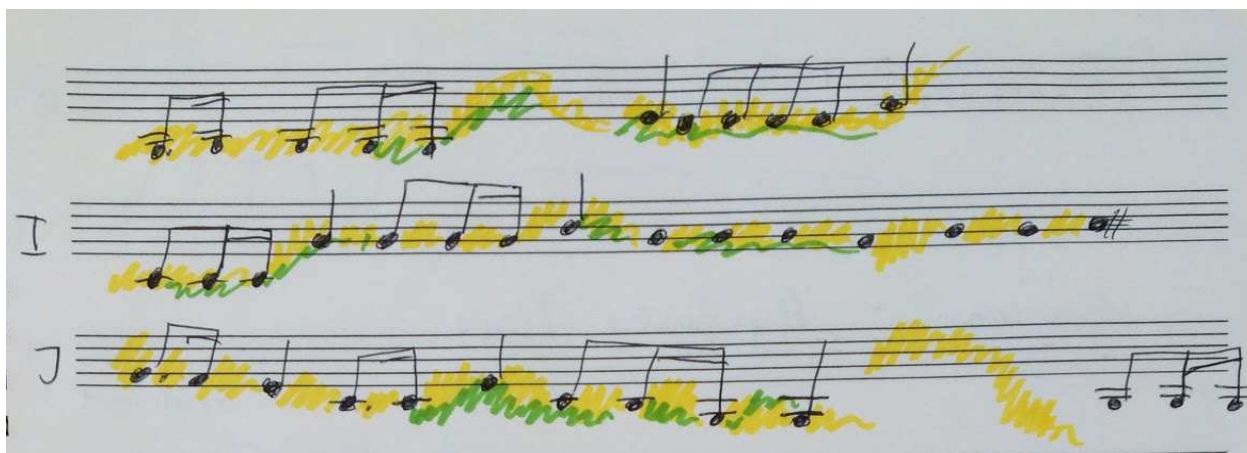
B⁴

C^u

5:48

ARCHIVES
125-12 Stave

Handwritten musical score for guitar, featuring 12 staves of music. The staves are labeled with letters B, C, D, E, F, G, and H. The music is written in a single system, with each staff containing a series of notes and rests. The notes are primarily eighth and sixteenth notes, often beamed together. The staves are decorated with yellow and green highlights, suggesting a specific playing technique or a particular sound. The word "ruffled" is written in the fifth staff, indicating a specific playing technique. The word "ARCHIVES" is visible in the bottom left corner.



Figures I, II, III, IV & V: Transcription of the Benjamin Ochavano Icaro

A ¹	B ¹	A ²	B ²	C	A ³	B ³
C ²	A ⁴	B ⁴	A ⁵	B ⁵	C [*]	D
A ⁶	B ⁶	C ³	A ⁷	B ⁷	C ⁴	

Figure VI: Section I structure, written out as 'phrases'. Colouring represents a 'group of phrases'³⁰.

A	B	C	B ²	A ²	B ³	C ²
B ⁴	C ³	B ⁵	C [*]	D	E	F
G	H	I	J			

Figure VII: Section II structure, written out as 'phrases'.

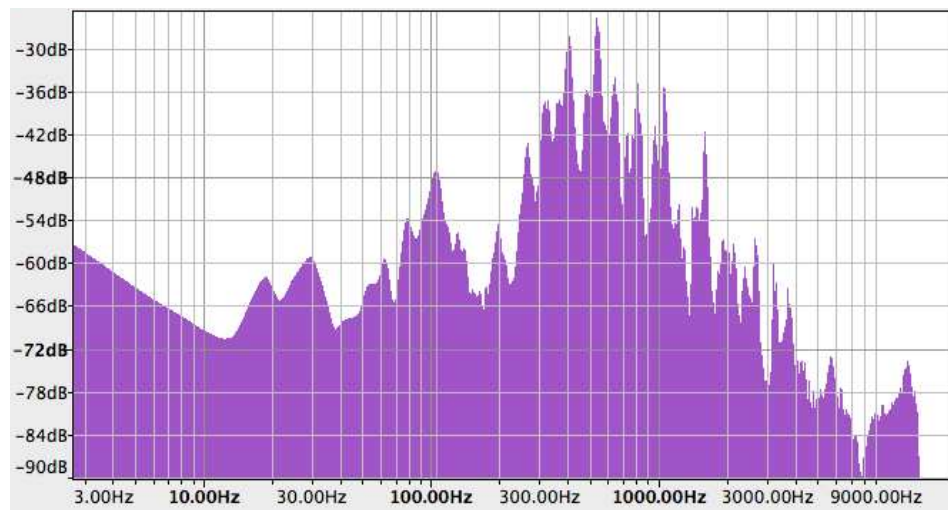
³⁰ All phrases marked red are recognised to be improvised repetitions of the same phrase, therefore sounding different but strongly related to other phrases marked red. The superscript numbering that accompanies the phrase names (A, B, C etc.) in Figure VI & VII, attempt to describe this evolving nature of the melodies.

3.1.1 - Notes on structure

In Section I, I have defined the most significant phrases as *A* and *B*, as they make up the majority of the section. Phrase *C* shares similarities in rhythm, but has significantly different spectral content. *C** is very similar to *C*, with a slightly altered ending. *D* is a unique phrase, featuring the only non-diatonic melodic content.

The second half (Section II) starts out in a similar way, with unique phrases *A*, *B* and *C*. *A* is very similar to *B*, differing in the opening phrasing. As there is only one use of *A*, perhaps it could be suggested that it is actually another variation of the phrase dubbed *B*. At approximately half way, the Shaman uses a phrase *C**, which is similar to *C*, differing only in the ending. What follows is a breaking of the repetitive, cyclic structure, with the shaman chanting seven unique phrases named *D*, *E*, *F*, *G*, *H*, *I* & *J*. Phrase *F* has been coloured red to mark its similarity with phrase *C/C**.

3.1.2 - Melodic Content

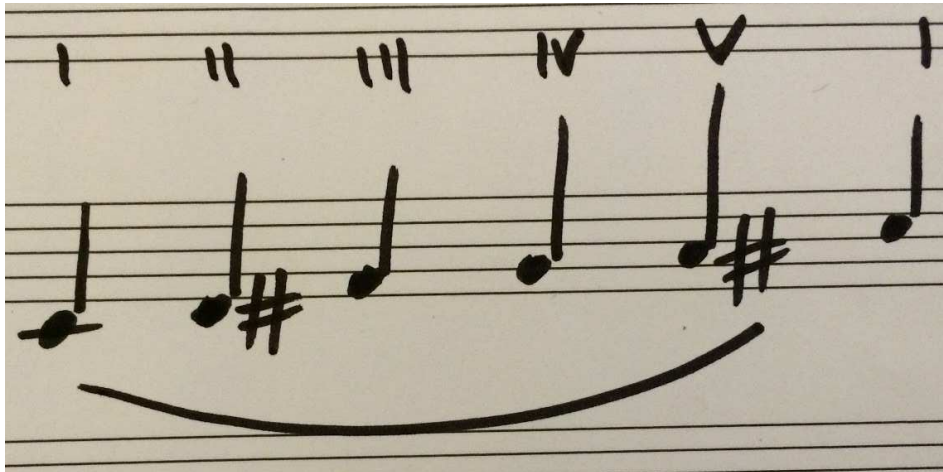


[Figure VIII](#): Frequency Spectrum for Opening 9 Phrases - X axis is Frequency (Hz), Y Axis is Volume (dB)

Figure VIII shows the frequency spectrum of the opening 9 phrases of Section A, which, as shown in Figure VI is comprised of phrases *A*, *B* and *C*. Immediately, an accentuation of the upper-middle frequencies can be observed, with 300 Hz-2000 Hz being

the most active area. The majority of the low-end frequencies are accounting for the background noise and the percussive instruments. The graph also shows the dominant frequencies being created by the chanting in the form of ‘peaks’. There are a significant peaks near 30 Hz, 110 Hz, 260 Hz, 420 Hz and 540 Hz. These frequencies translate as pitches **B** (borderline **C**), **A**, **C**, **A**³¹ and **C**. Both the *frequencies* (in Hz) and the **pitches** are approximations, as each frequency ‘peak’ on Figure VIII represents a band of frequencies, the most accentuated of which is the dominant frequency (referring to the frequency that is *most present* and therefore becomes its classification). The *most present* frequencies suggest that the song is harmonically rooted in **C**.

Continuing further, the score demonstrates the use of notes which correspond to, what appears to be a diatonic pentatonic scale rooted in **C**; aligning with the frequencies found in Figure VIII. All the changing melodies are constructed from these pitches, except one deviation found in phrase *D*. This root of **C** shifts gradually throughout the performance however, and this is mentioned in [3.1.4](#).



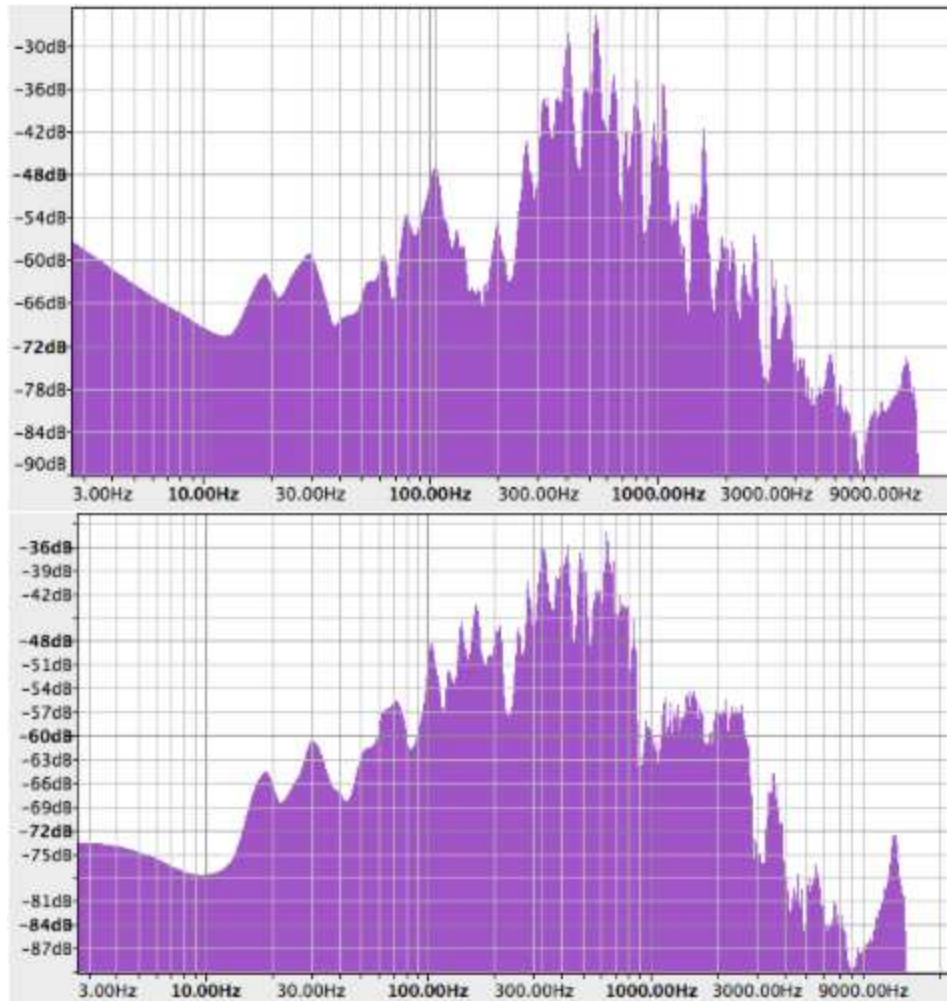
[Figure IX](#): Pentatonic scale observed in Benjamin Ochavano Icaro.

Phrase *D* contains the only part in the Icaro that deviates from the diatonic structure, exhibiting use of pitch **G** $\flat$ at a point in the song where the 5-note scale is not including this note. In section II there are changes in rhythm and pulse, but the melodic content remains similar; the section continues to follow the **C/C#**-rooted pentatonic that preceded the transition between sections.

As the scores (Figures I, II, III, IV & V) depicts in yellow, the general melodic movement tends to follow an undulating pattern. In Section I, phrases *A* and *B*, show a

³¹ The peak is less clear as to which pitch it is describing.

movement of ‘*down, up, down*’, while phrase *C* exhibits movement of ‘*up, down, up*’. Phrase *C* differs largely from *A* & *B* (in terms of melodic content), as it is based on the dominant note³² of the scale *G/G#*. The transcriptions were made aurally using a pitch reference, and the frequency spectrum adds more evidence to support their accuracy in capturing the essence of the music.



[Figure X](#): Spectral analysis of the last 7 phrases of the Benjamin Ochavano Icaro

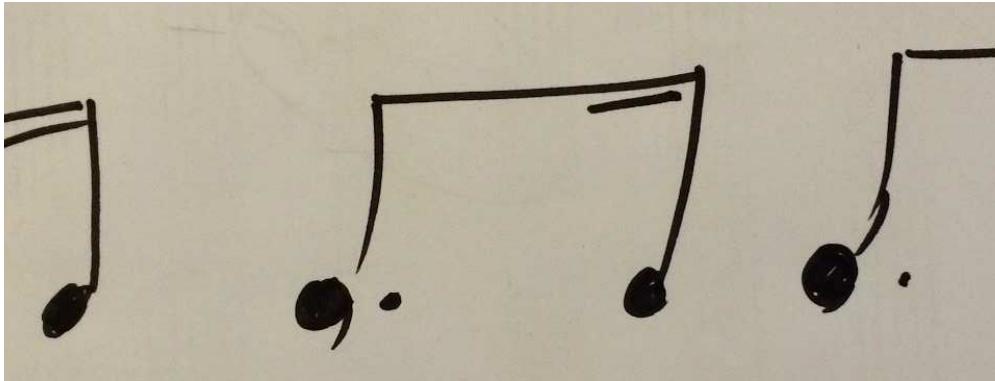
As described in [3.1.4](#), the song drifts upwards in pitch, eventually reaching a tonic of *C#*. At this point, despite the change in ‘key’ or *root note* the five note scale structure stays the same. Figure X shows a comparison between the frequency spectrum of the opening phrases to the frequency spectrum for the closing phrases. What can be clearly observed is movement from a higher register, to a slightly lower one defined by an

³² A western term that regards cadences (intervallic relationships).

increase in the lower middle frequencies. Section II appears to make more use of lower frequencies.

3.1.3 - Rhythmic Content:

There is a heavily-swung feel to the rhythm (resulting in a heart-beat-like pulse) which has been notated on the score as a '*dotted quaver - semi-quaver*' rhythm. Figure XI depicts this dotted rhythm, as this pattern stands out as a significant rhythmic motif.



[Figure XI](#): Example of the '*dotted quaver - semi-quaver*' rhythm.

However, each 'phrase' is disconnected from each other by irregular pauses, and these irregularities intervene with accurately determining more than an empirical grasp on the pulse, or BPM (beats per minute). Many of the phrases fade-out towards the end³³, which results in further loosening of this rhythm. In light of these complications, the general pulse of Section I has been found to be around 103 pulses per minute, however, the strong sense of flexibility, variation and loose rhythmic strictness could be interpreted as a dominant, therefore significant, rhythmic motif worthy of explanation in the Further discussions of [Chapter 5](#). The method ultimately used for finding this pulse involved using 'Logic Pro X'³⁴ to compare fragments of the track to a timing guide³⁵. By taking loops of individual phrases and overlaying them onto the metronome, a sense of pulse could be more accurately determined.

In Section II, there is a clear change of pace, and a more determinable rhythm. There is an apparent lyrical change which also contributes to the bouncier rhythm of the

³³ Due to mumbling and lack of clarity in the chanting towards the end of every phrase.

³⁴ An industry-standard Digital Audio Workstation, specialised in handling and manipulating audio.

³⁵ In this case a metronome and the sequencer grid in Logic Pro X.

second movement³⁶. The beat has been determined to be approximately 128 pulses per minute, although the singer seems to slow down quite considerably as the pitch descends.

As mentioned in the Methodology, there is an underlying issue found in lack of visual accompaniment making it difficult to tell whether the seemingly dramatic change in rhythm and pulse aligns with a new, inter-ceremonial phase. Perhaps the speeding up is the shaman now employing a new 'Icaro' from his 'collection', a 'technique of ecstasy' as described in sections [1.6](#), and [4.4](#) (Eliade, 1964).

3.1.4 - Harmonic Content:

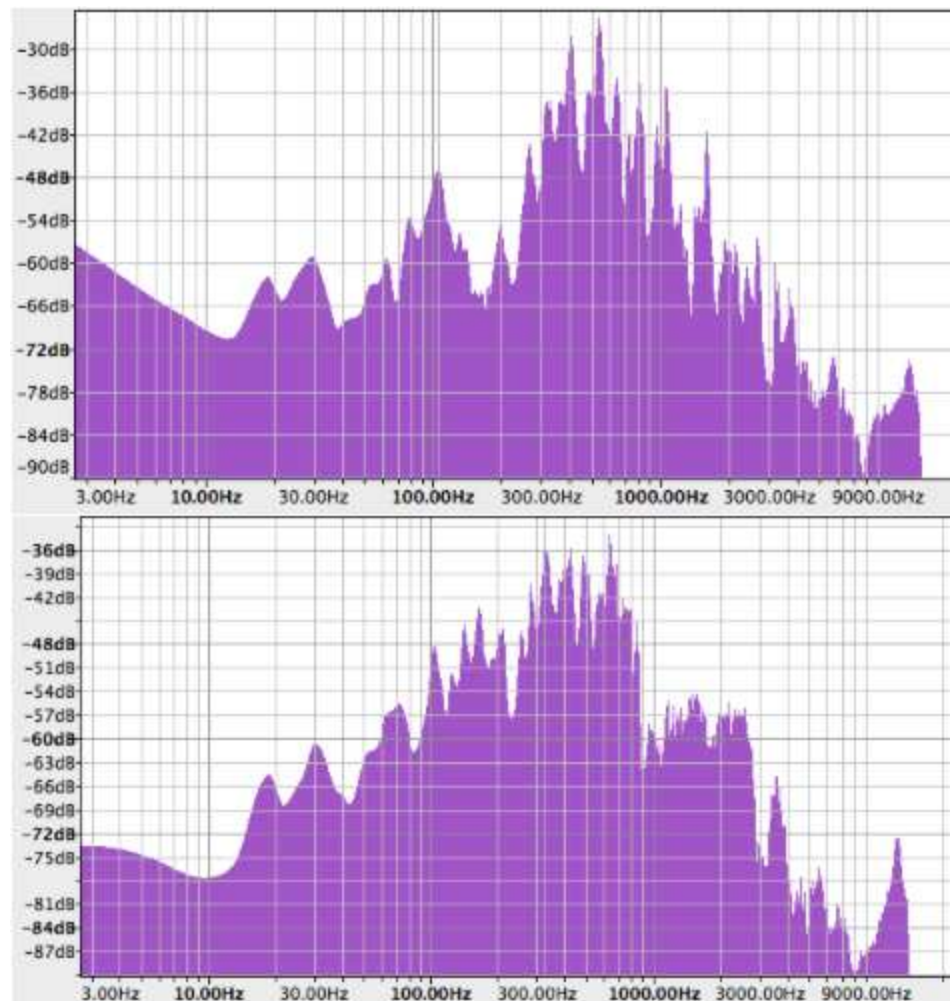


Figure XII: Comparison of opening and closing periods of the recording of Benjamin Ochavano's Icaro

³⁶ The phonetics sound more rounded and rhythmic with the lyrical changes in place.

When notating the harmony, one intention was to identify when and where, and in how much quantity, there is use of secondary voices or instruments. The general pattern found when observing this was that the harmonic content of Section I (notated in green in Figures I-V) there appeared to be harmonic accentuation around the middle of each phrase.

These changes in volume could be caused by an issue outlined in the methodology, whereby movement within the ceremonial grounds can affect the recording. However, due to lack of stereophonic movement it can be suggested that the harmony singer stayed in a similar position. This second voice quietly hums a detuned, simplified imitation of the Icaro, appearing to follow the lead. The second voice avoids making sound at the very start of the phrase. The detuned voice creates a dissonance when blending with the lead, this could be a sound tactic in generating *binaural beating*; another shamanic technique (Eliade, 1964) for altering or manipulating ‘ecstasy’.

A second interesting aspect of Harmony that can be discussed in this Icaro is the shift in the ‘root’. As supported by the frequencies in Figure XII, by the end of the Icaro, the harmonic root has shifted upwards to C#. It is difficult to speculate on whether this is an intentional shift or not. If the shift is intentional, then there is reason to speculate upon the effects of shifting harmony in music and/or psychedelic experience. If the shift is not intentional, then there is reason to investigate the lack of strictness in keeping the ‘tonic’ consistent. If, as Luna states, the melody contains the *healing power*, a lack of strictness in which pitch is being generated or sung could indicate that the intervallic qualities³⁷ of each consecutive note could potentially be more significant than specific pitches.

3.1.5 - Other observations:

After phrase *B'*, there is a fairly long pause for tobacco blowing³⁸ which is not repeated again in the Icaro. There are shorter pauses that feel more rhythmic, and these more frequent pauses do not appear to relate to the long pause that occurs after Icaro *B'*. This break, being at an irregular time, suggests that this recording may have been taken midway through a ceremony. As these recordings were made in-situ (Charing, 2015), the irregular timing of brief interlude can suggest that the recording is potentially not a complete musical movement, but an extract of a much longer performance.

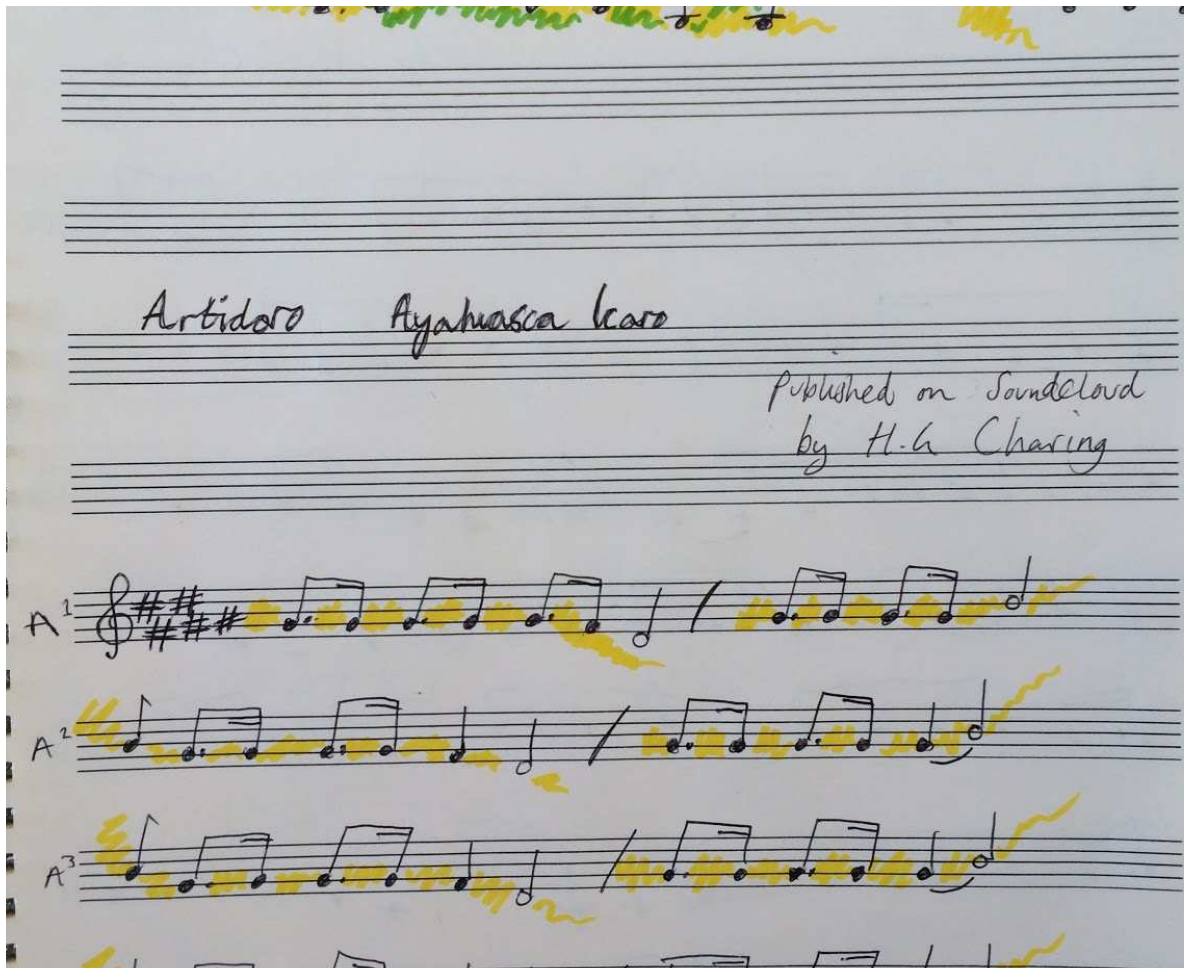
³⁷ The rhythmic and spectral distance between each note.

³⁸ This can be referred to as ‘Ikaray’, the word for ‘blow smoke’ in the Quencha language.

3.2 - Artidoro Ayahuasca Shaman

Published by H.G Charing on Soundcloud: www.soundcloud.com/hgcharing

Track Length: 4:07



Figures XIII & XIV: The transcription of Artidoro's Icaro

3.2.1 - Structural Content:

A	B	C	D	E	F
A	B	C	D	E	F
A	B	C	D	E	F
A	B	C	D	E	F
A	B	C	D	E	F

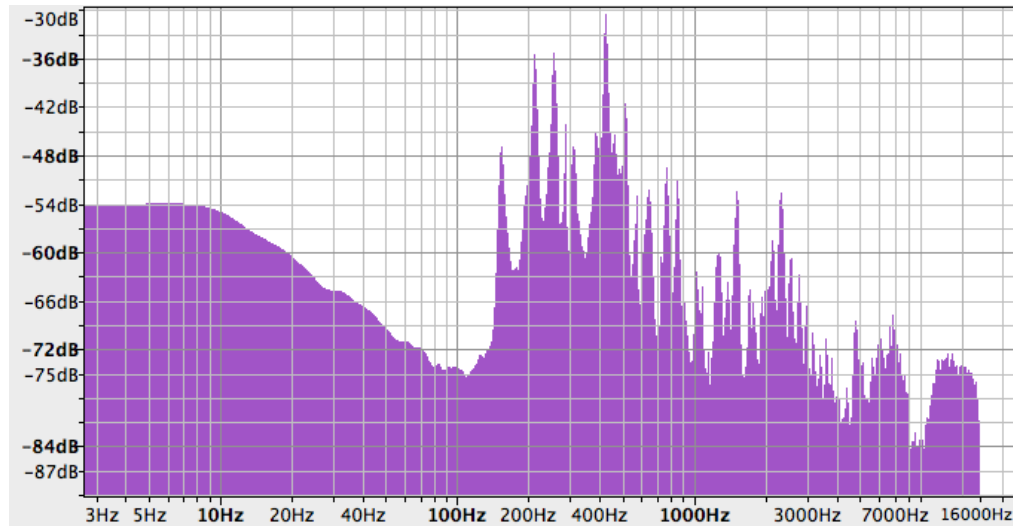
[Figure XV](#): The structure of Artidoro's Icaro presented as a chart.

The structure of this song can be described as five cycles of six phrases. This is shown in Figure XV as five iterations of phrases *A-F*. Phrases *A-E* are very similar, with subtle variation. However phrase *F* is shorter, containing a resolving quality in the rhythm and melody, forming the end of the cycle.

The subtle variations in the phrases can be inferred as an attempt to develop the music in a progressive manner while maintaining a specific structure which would indicate significance in the structure of Icaros.

3.2.2 - Melodic Content:

This Icaro demonstrates a minimal amount of melodic variation, with very simple phrasing that doesn't change over the performance. With such repetitive phrasing, the majority of spectral variance stems from the natural timbral shaping produced by language. While the modulating lyrics do subtly affect the melodic nature of the Icaro in this way, It is unclear how much these timbral changes impact and contribute to the success or purpose of the song. Cultural literature (Luna, 1986) suggests great importance in the melody, but the lack of study or discussion on lyricism doesn't prove unimportance.



[Figure XVI: Frequency Spectrum for Artidoro's Ayahuasca Icaro](#)

The spectrum in Figure XVI shows the spectral content in the opening minutes of Artidoro's chanting. There are dominant peaks are 153 Hz, 209 Hz, 252 Hz and 417 Hz³⁹, which translate approximately as **D#**, **G#**, **B** and **A#**. Each phrase consists of a long chain of **G#** notes, dropping to **D#** in the middle, and rising to **B** at the end of each phrase. This forms an interesting pattern of repetitively rising and falling. Occasionally, the Shaman will commence a phrase with by swinging up or down by a single step of the scale before the first note.

At the end of each phrase denoted *A* there is a sharp rise in pitch upon the final note, which is sustained for multiple beats until it has completely faded out. *B* significantly differs from this melodic upswing, remaining straight on the note **G#**. This has been depicted on the score as a crotchet - minim connected with a tie, because the final **B** is more of a slide upwards than a defined note.

3.2.3 - Rhythmic Content:

Considering the heavy repetition of a simple, slightly monotonous⁴⁰ melodic motif, it can be suggested that, in this Icaro, the rhythmic content is more prominent feature. Matching the Benjamin Ochavano Icaro, Artidoro's melody is based on a *dotted*

³⁹ It is an interesting side note that the dominant frequency here (417 Hz), is a member of the Solfeggio frequencies. To remain brief, as speculating upon this archaic concept is far beyond the scope of this research, these frequencies are dominant in both Gregorian chanting and Ancient Indian music, and are suggested to be 'healing frequencies'. (Hulse, 2009)

⁴⁰ In that it remains on a **G#** for a significant amount of time.

quaver - semi-quaver rhythm, identical to that in Figure XI⁴¹. It continues with this rhythm throughout the entirety of the song. The last phrase of the cycle, *F*, ends with a change of rhythm, which grants the phrase a resolving quality, marking the end of the cycle.

3.2.4 - Harmonic Content

The predominant interval structure, or scale, is based on what is analogous to a **C#** pentatonic scale⁴², although the heavy focus on the note **G#** could suggest otherwise. If there is a relation to the **C#**, then this Icaro is the second Icaro to be linked to the harmonic frequencies of **C/C#**. However, overall there appears to be little harmonic interest in this Icaro.

⁴¹ As can be seen in the score in phrase *A-F*.

⁴² In that, the notes used form a scale comprised of the same notes as a **C#** major pentatonic, only the focus is on the Dominant 5th **G#**.

3.3 - Mariana Shaman Icaro

Published by H.G Charing on Soundcloud: www.soundcloud.com/hgcharing

Track Length: 8:38

Mariano - Ayahuasca Icaro

published on SoundCloud
by H.G Charing

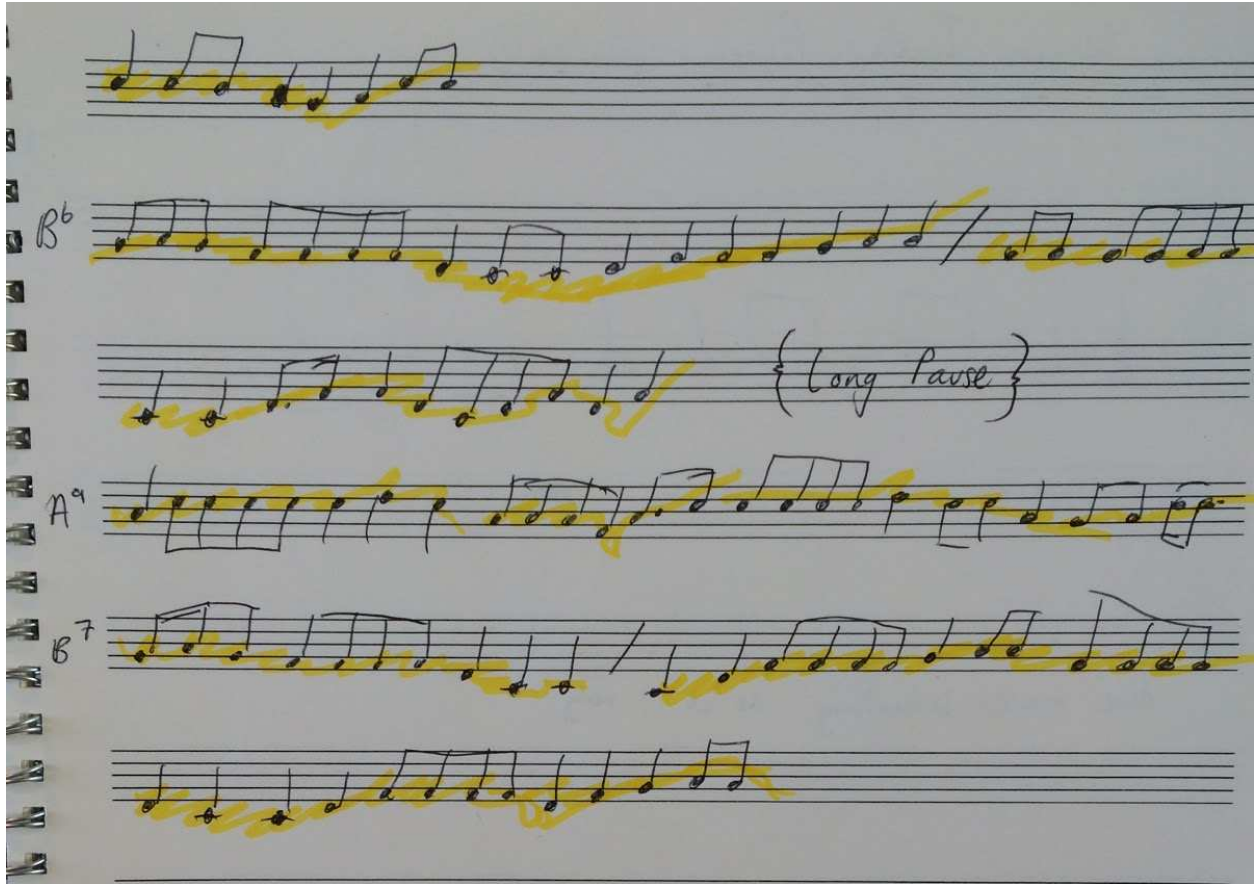
The image shows a handwritten musical score on ten staves. The title 'Mariano - Ayahuasca Icaro' is written at the top left, and 'published on SoundCloud by H.G Charing' is written at the top right. The score begins with a treble clef and a key signature of three sharps (F#, C#, G#). The music is written in a melodic style with various note values, including eighth and sixteenth notes, and rests. The score is divided into sections labeled A, B, and A². Section A spans the first two staves, section B spans the third and fourth staves, and section A² spans the fifth and sixth staves. The notation is handwritten and includes some yellow highlighting under the notes. The staves are numbered 1 through 10 at the bottom left.

Handwritten musical notation on a spiral-bound notebook page, featuring five systems of staves. The notation is written in black ink, with yellow highlighting under the notes. The systems are labeled on the left:

- A³**: Two staves of music.
- B²**: Two staves of music.
- A⁴**: Two staves of music.
- B³**: Two staves of music.
- A⁵**: Three staves of music.
- B⁴**: One staff of music.

The notation consists of eighth and sixteenth notes, often beamed together, with some rests and bar lines. The yellow highlighting is applied to the notes and stems, following the melodic lines.

A handwritten musical score on ten staves. The notation includes various note values, rests, and bar lines. Several staves have a wavy yellow highlight underneath the notes. Annotations include:
 - Staff 1: A bracketed note with the text "(long pause)" written above it.
 - Staff 2: A handwritten "x" at the beginning and an "A⁶" at the end.
 - Staff 6: A bracketed note with the text "(long pause)" written above it.
 - Staff 7: A handwritten "A⁷" at the end.
 - Staff 8: A handwritten "B⁵" at the end.
 - Staff 10: A handwritten "A⁸" at the end.
 In the bottom left corner, the word "ARCHIVES" is printed above the number "125-12 Stave".



Figures XVII, XVIII, XIX, XX: The transcription of Mariano's Ayahuasca Icaro.

3.3.1 - Structural Notes:

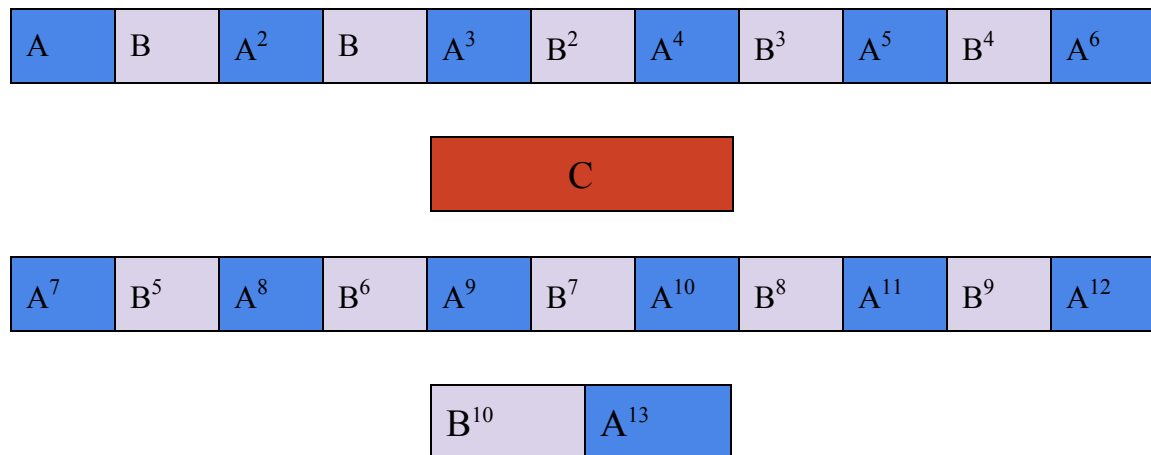
A	B	A ²	B	A ³	B ²	A ⁴
B ³	A ⁵	B ⁴	A ⁶	C	A ⁷	B ⁵
A ⁸	B ⁶	A ⁹	B ⁷	A ¹⁰	B ⁸	A ¹¹
B ⁹	A ¹²	B ¹⁰	A ¹³			

Figure XXI: Structure of the phrases in Mariano's Icaro.

In Mariano's Icaro, there is a cycle of two phrase archetypes, denoted *A* and *B*, and, as indicated by the superscript numbers, each phrase changes slightly each repetition. As Mariano cycles through these repetitions, the phrases become longer and more complex. Despite the developing nature, the two 'voices', 'A', and 'B', remain distinct enough that they are audibly recognisable as being related to the previous

iteration of that archetype⁴³, and because of this the structure of this Icaro can be defined as a call and response pattern. It appears as a conversational structure, in which the phrases denoted as *A*, act as voice one, and appears to lead, with the response coming in the form of phrase *B* and its variations.

There is a counterpoint phrase in which the song appears to turn around, which has been labelled *C*. It stands out from either voices with a start which is not seen in the rest of the Icaro, yet, as the score reveals, the phrase *C* features an ending that is very similar to the phrase *B*.



[Figure XXII](#): An alternative demonstration of the structure, attempting to show a pattern

Figure XXII shows an alternative way of displaying the structure. Here, the structure can be shown as a palindrome, using *C* as a turning point. Due to the limitations discussed in [3.1.1](#) and [3.2.1](#), as , it's difficult to tell whether the piece commenced before the recording, and whether the recording captures the whole Icaro. Should the performance have gone beyond what the recording captured, this structure does no longer appear to be as symmetrical.

⁴³ All *A* phrases are recognisable as an *A* phrase, despite the fact they are gradually changing, and the same applies to all *B* phrases.

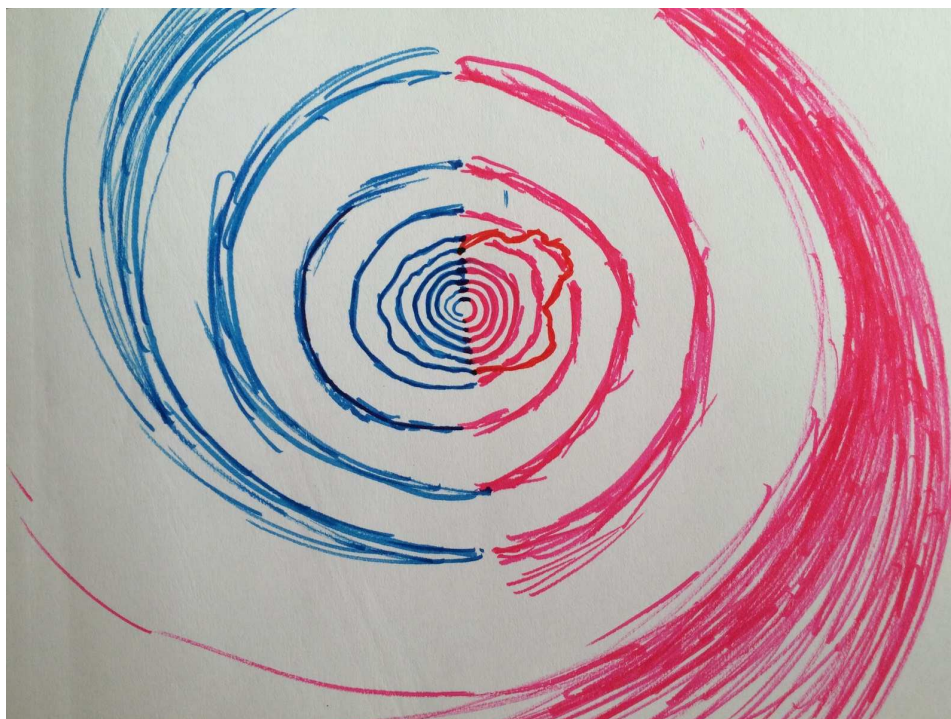


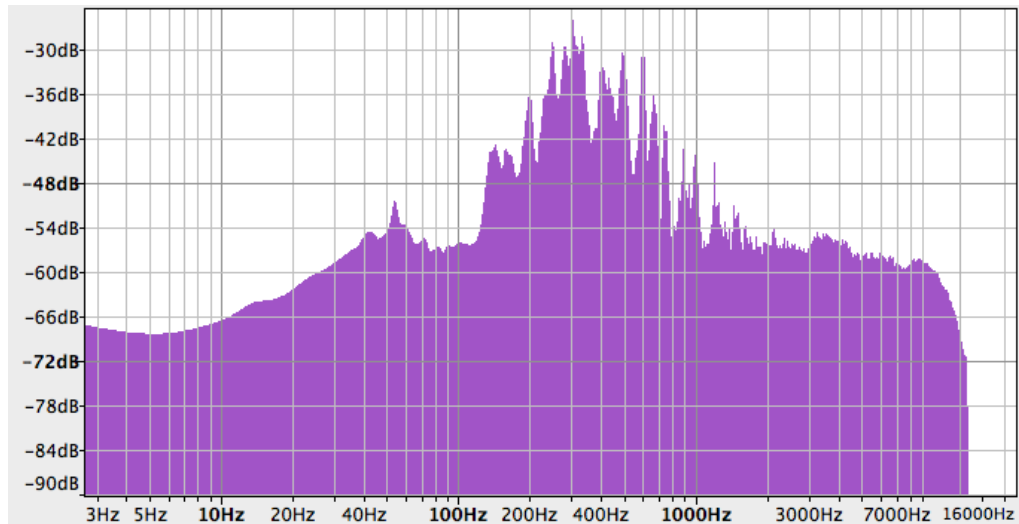
Figure XXIII: An abstract transcription of Mariano's Ayahuasca Icaro

A third alternative score displays the Icaro as a spiral, and this more aptly demonstrates the modulating nature of the Icaro. As the spiral expands from the centre, the length and complexity increases, in the same way, each cycle gets more complex and longer. The break from the balanced cycle, depicted as an awkward, extended line in red is the *C* phrase near the middle.

3.3.2 - Melodic Content:

As suggested in the Structural Content section, the two main melody lines, *A* and *B*, evolve as the song goes on. Each phrase differs slightly in melodic rhythm⁴⁴, following a vague melodic pathway, with no more than 3 or 4 minor changes per cycle. However, , by the 6th repetition, *A* is completely different to what it started out as. This evolving quality substantiates previous findings regarding variation, in that, all Icaros seem to directly avoid straight repetition. This could be a lack of performance ability, or a loose strictness towards rhythm, or another example of variation.

⁴⁴ The timing interval between each note in the melody, rather than the grid or structure which the melody is overlayed (coming in the form of the Schacapa rattling in this Icaro).



[Figure XXIV: Frequency Spectrum for Mariano's Ayahuasca Icaro](#)

Once again, the tonic is centred on the **C-C#** region, using the same pentatonic scale structure as the other pieces. This is shown in figure XXIV, with prominent peaks at 142 Hz (**C#**), 427 Hz (**G#**) and 484 Hz (**B**). The frequency spectrum (figure XXIV) shows that there is also a leniency towards the middle frequencies throughout the song. Considering the lower frequencies consist of the rattling and background noise, it's suggestible that the vocals consist of the middle and high frequencies, coinciding with the other Icaros.

3.3.3 - Rhythmic Content:

What immediately stands out about this piece is the utilisation of a strong pulse kept in constant motion with a rattle. This pulse (depicted in Figure XXV) is kept through the entire duration in a comparably strict rhythm, not stopping with any vocal pauses. It is only at the end where the rattling stops and the shaman repeats the icaro once alone. The pulse of the Shacapa is approximately 105 beats per minute.

[Figure XXV: The pulse of the Schacapa in Mariano's Ayahuasca Icaro. Steady Quavers.](#)

3.3.4: Harmonic Content

Mariano's Icaro doesn't exhibit any signs of a secondary voice, or any facets of interest in the melodic harmony, other than the limited nature. In this sense, the harmonic content is similar to that of Artidoro's Icaro (3.2).

3.4 - Elf's Bow (Extract)

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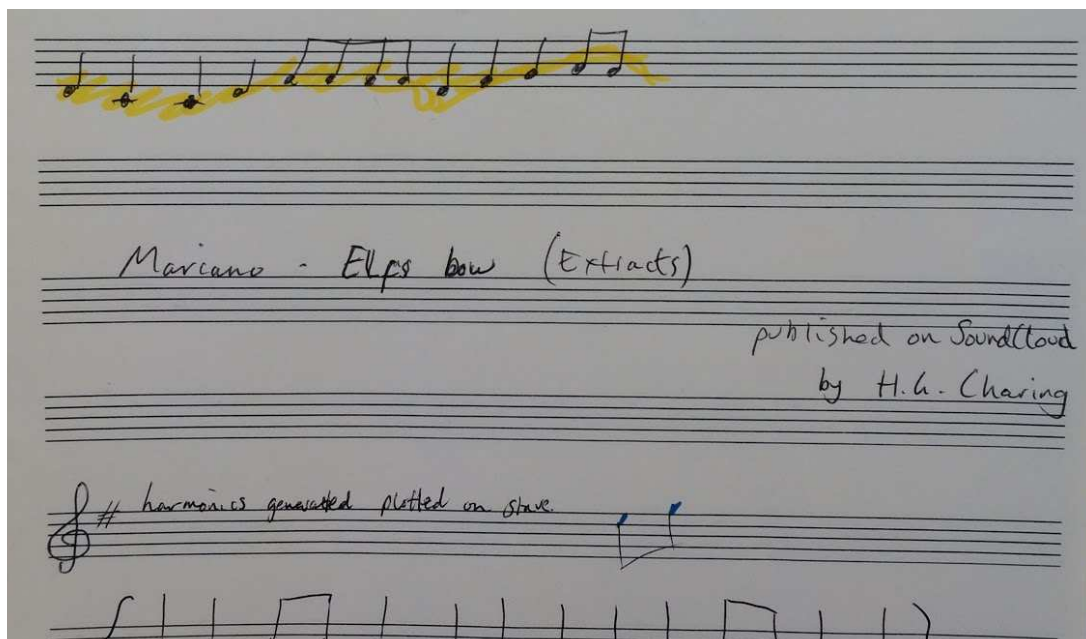


Figure XXVI: A transcription of a brief extract from the beginning of Mariano's 'Elf's bow' Icaro.

Rigid straight pulse that makes minimal rhythmic changes every so often, as if trying to catch the listener off guard. The instrument named 'Arc de Duende' (Elf's bow), described by Charing (2014) as a 'magical stringed instrument', is quite likely to be similar to a Berimbau. This instrument is strongly associated with Capoeira dancing (Downey, 2002). Downey suggests that Capoeira instrumentalists describe the experiential effects of sound textures, especially those created by the Berimbau, in such a way that has proven difficult for ethnomusicologists to grasp in the past.

The melody that resonates off this instrument is subtle, very faint changes in pitch. The timbre of the instruments resonance is similar to a formant filter sound. In this sense, the Elf's bow creates a human-vocal quality. The methodology is quite insufficient for dealing with this

music, and an alternative has been devised to at least demonstrate some of the musical qualities, as this is too, a healing song.

What has been depicted in the score (Figure XXVI) is a steady crotchet pulse with melodic movements overlayed. The selected extracts contain a general sense of how the melody moves upwards and downwards in steps around the central **B** note⁴⁵. It is possible that, due to the instrument and how it is best played to produce sound, constant step variation is required to exhibit a sense of rhythmic melody.

Spirit Quest recordings

3.5 - Don Rober Recordings

Published on SpiritQuest: Available at: <http://biopark.org/icaros.html>

A series of recordings available from the *SpiritQuest* centre includes a folder entitled ‘*Rober recordings*’.⁴⁶ There are 4 pairs of recordings which could possibly showcase the differentiating Icaros of one long Ayahuasca ceremony, conducted by *Don Rober Acho Jarama*. In these short recordings, a singing and whistling style is employed in a similar way to the previous recordings that have been examined so far. Each of the ‘*Rober*’ extracts contain a pulse played by a *schacapa*.⁴⁷ These recordings also demonstrate similar use of harmony, in that, one backing singer, perhaps the apprentice or an assistant to the shaman, intermittently applies a dissonant harmony to the lead voice, layering the sound into an even more spectrally complex, polyphonic sound.

SpiritQuest bring extra attention to the high level of experience held by the performer here (Don Rober), noted to be the incomparable maestro. Ayahuasca shaman authenticity, and their ‘power’ is suggested to be based in their repertoire of icaros and their knowledge of how to employ them ritualistically. Having access to the recordings of a maestro is useful in identifying which other icaros are coherent with the, almost platonic, Ayahuasca Icaro (Luna, 1992).

3.6 - Hermana Maria Luisa Tuesta Flores.

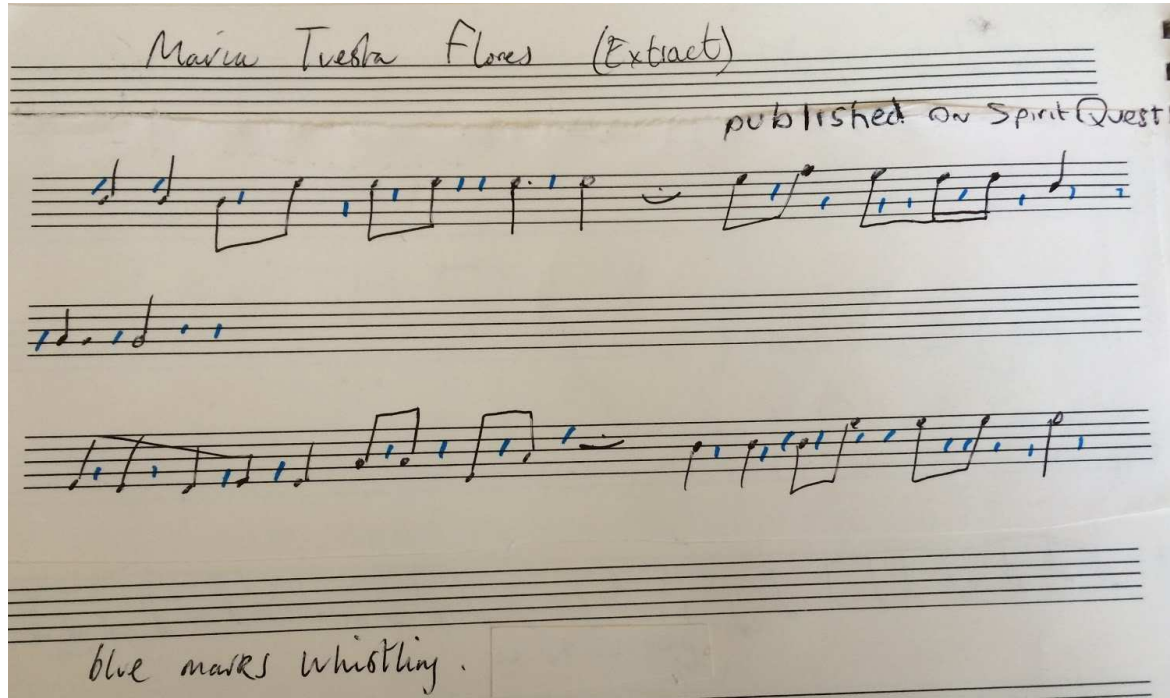
Upon inspecting the vocal performances of this adept, there is an immediate factor which separates it from the others. In the recordings filed under her name, Maria’s icaros appear to have

⁴⁵ The main pitch of the instrument is a low **B** note created by hitting the string in its primary position.

⁴⁶ This refers to Don Rober Acho Jarama, an ayahuasquero involved with the *Spirit Quest* centre.

⁴⁷ Similarly to the Huitoto shaman Mariano

more than one performer overlapping multiple melodies. While she remains the dominant voice, there is a second voice present (in the form of whistling), both accompanied with a Schacapa rattle playing a straight rhythm.

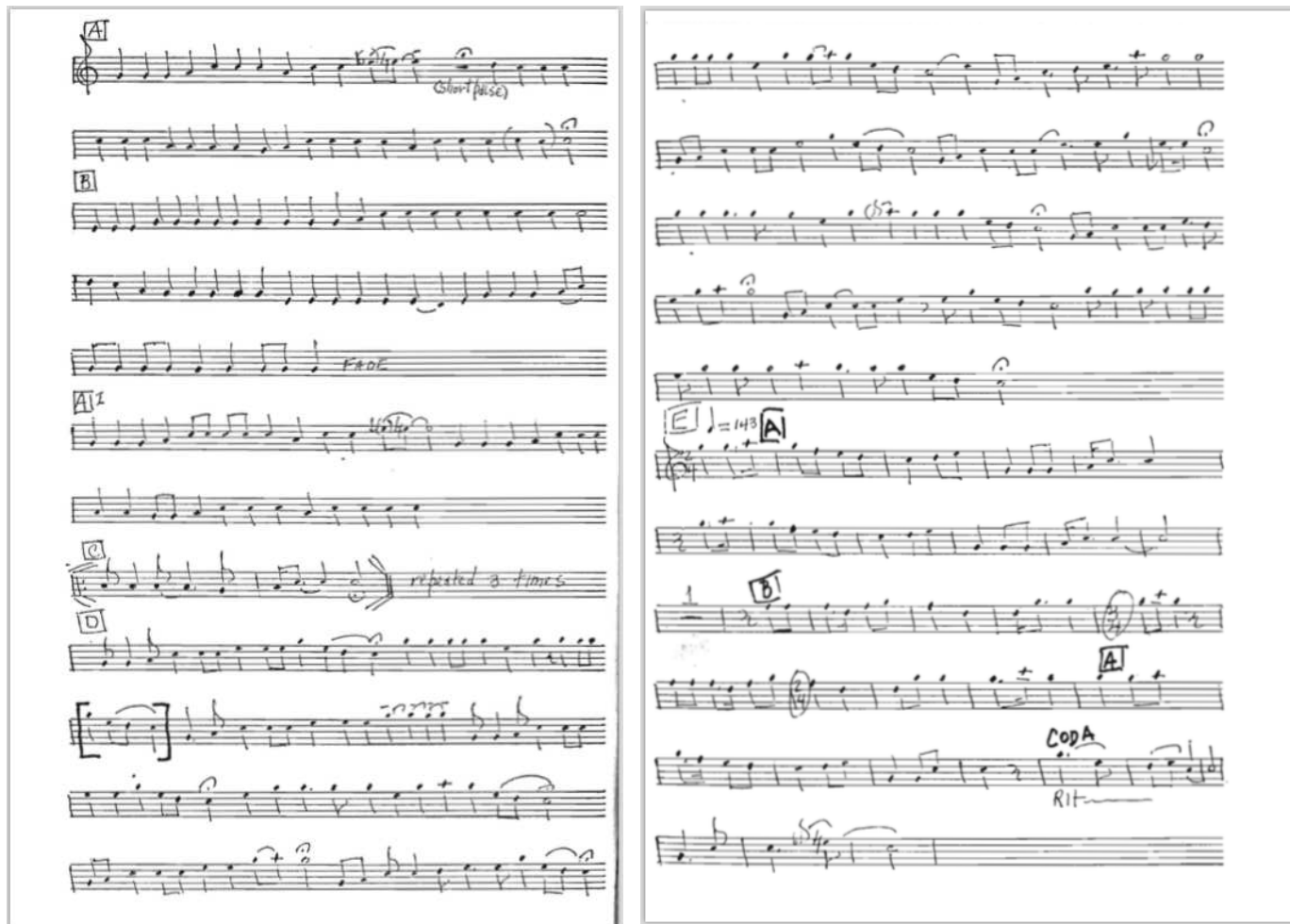


[Figure XXVII](#): The transcription of a short extract from the beginning of Maria Tuesta Flores' Ayahuasca Icaro

The whistling and the chanting have no rhythmic connection, with the chanting following the rhythm of the shacapa, and the whistling appearing to operate totally independently. With this in mind, the whistling has been treated as the dissonant harmony that usually accompanies Icaro chanting. Markings on the score extract show how the harmony part appears to cease during times that the main voice falls to a slightly lower register.

3.7 - Marlene Dobkin de Rios score

This score is available in the paper 'Hallucinogenic Music' by Fred Katz and M. Dobkin de Rios (1971). The score describes an Icaro that the pair had recorded during their time in Peru. It stands as a valuable information source to compare to my own scores.



[Figures XXVIII & XXIX](#): The transcription produced by Marlene Dobkin de Rios (1971)

3.7.1 - Structure

The structure can be written out as *A, B, A - C, D, E - A, B, A*. This structure, if it is the entire length of that particular Icaro, appears to be another instance of structural symmetry. When written out this way, it also appears to be more of a strophic structure than other Icaros, and could be written as Verse - Chorus - Verse.

3.7.2 - Melody

The score appears to be suggesting a tuning of C major/A minor (no sharps or flats), and shows a dominant use of the note ‘C’. This key signature doesn’t change throughout, suggesting limited shifting of pitch⁴⁸. The transcriber also notes

3.7.3 - Rhythm

Dobkin de Rios describes a changing metre throughout the song, a feature that hasn’t been observed in the scores produced in response to the HG Charing recordings, perhaps highlighting areas in which the any further transcriptions can be improved. The transcription notes a few instances of the dotted-quaver rhythm⁴⁹, but it does appear to be a prominent rhythm throughout, featuring only in four phrases. At 143 pulses per minute, the tempo is also higher than other Icaros that have been analysed here.

⁴⁸ Such as the pitch shifting observed in Benjamin Ochavano Icaro.

⁴⁹ A recurrent rhythm throughout the Icaros.

Chapter 4

Discussions

4.1 - Melody

In a large majority of pieces analysed, including the score of Marlene Dobkin de Rios (Figures XXVIII & XXIX, 1971), there is a central melodic theme, best described as a diatonic scale comprised of 5 unique pitches. This contrasts the expectations in the methodology, whereby western notation proved quite appropriate to describe the movement in pitch. It is interesting to note that many of the Icaros analysed also root this 5-note scale on a pitch between C and C#. The ‘Benjamin Ochavano Icaro’ piece starts in the key of C, but as described in [3.1](#), slowly shifts upwards until it no longer resolves to a C tonic, but C#. Due to the significant use of this pitch, further research could be done into the relevance of certain frequencies.

An interesting observation is that, these five-note scale structures, such as that depicted in figure IX, use intervals that appear analogous to a pentatonic scale that is also found in many other cultures. It must be noted that little allowance for any microtones has been made, as it is difficult to say for certain whether there is deliberate detuning, accidental detuning, or even accidental microtonal inaccuracies.

It is proposed by the author that, melodic movements of up and down⁵⁰, have a particular phenomenological perception (for example ‘soothing’) which in turn produces a change in consciousness for the participant. The literature suggests that the Shaman deliberately employs Icaros as a ‘tactical’ method of manipulating consciousness (Molina, 2009; Eliade, 1964; McKenna, 1992), and inducing phenomena in perception could be an example of this.

4.2 - Rhythm

Multiple Icaros analysed ([3.1](#), [3.2](#), [3.3](#)) make use of a dotted crotchet rhythm, either in a dominant way ([3.1](#), [3.2](#)), or less dominant but still highly utilised ([3.3](#)). This suggests there is a

⁵⁰ Such as that in Benjamin Ochavano phrases A & B.

significance to this rhythm within the Ayahuasca ceremony, and therefore perhaps has a significant effect on the brain or perception.

Many of the Icaros analysed ([3.3](#), [3.5](#), [3.6](#)) included the thematic use of Schacapa rattling. This straight rhythm creates a strong sense of pulse for the melody to be sung to. Icaros that included the Schacapa had a stricter sense of rhythm in the melody, as it can be used as a timing guidance. The Schacapa also generates a large array of frequencies making it a significant contributor to the overall harmony too.

4.3 - Harmony

There are different applications of Harmony presented in the Icaros, from little harmonic interest ([3.2](#)), to a dissonant second voice ([3.1](#)), to a totally overlapping pair of musical ideas ([3.6](#)).

The main role of the harmony part, across all applications of a second voice, is to give the melody a more complex frequency range. Furthermore, the main theme of the harmony created can be described as dissonant and counterpoint. The frequencies do not tend to match, with the dissonance stemming from the seemingly deliberate spectral contrast. As discussed in [2.5.3](#), the binaural beating caused by two almost identical frequencies clashing, could be the purpose of this dissonance. Binaural beats interact with the brain in a similar way to polyrhythmic drumming, specifically regarding *theta* brainwaves (Maas, et al., 2009; Atwater, 1997). It appears that these theta waves are closely connected to music and altered states of consciousness. The author suggests that the use of hallucinogenic substances, when relating to music, is to put the subject in the right state of mind for the music to take hold of, and control the frequency of the brain.

4.4 - Instrumentation

Anthropological literature on amazonian culture suggests that non-verbal sound is one of the specific ways to communicate with spirits. (Chaumeil, 2011b) This includes whistling, chanting, humming. Chaumeil goes on to say that Yagua shamans use certain exercises to enable singing in very high frequencies, which they say corresponds with the way the spirits speak. Chaumeil (2011b) cites Yvinec, who suggests that it is not ‘what is being communicated’ that matters, only that they ‘are communicating/connecting’; the very principle of engaging in an alternate process of communication. The Ayahuasca Icaros analysed have demonstrated a significant use of high frequencies, across the different instruments used to create them, which could be linked to the process of engaging in alternate communication methods exhibited by the Yagua.

It has been observed in these recordings that Ayahuasca Icaros include whistling, chanting, humming, Berimbau and a recurrent Shacapa. None of these featured a flute, which contradicts some literature (Katz, et al., 1971; Uzendoski et al., 2011). While not limited to, the dominant instrument type is wind instruments or more accurately, aerophones. ‘Burst of Breath’ (2011a) explores the many examples of aerophonic significance in many South American cultures. Further research could be conducted into the sonic characteristics unique to aerophones, as well as psychological studies of how aerophones are perceived differently to other instrument types. Chanting and vocalisations can be understood as containing nuances that are unique to the human voice, a sound the brain is incredibly adept at analysing for semantic information or subtle differences.

The use of Berimbau also exhibits a complex sound with varying resonances creating a formant effect, which in a sense liken it to the human voice in terms of versatility and complexity and timbre. Stringed instruments that vibrate in this way, as well as bowed instruments like the violin, create very complex sounds that could also influence the brain in a different, but equally intense way, thus worthy of further research as well. As observed in the recordings of Huitoto Shaman Mariano, Shamans use more than one instrument during a ceremony. Mariano’s (3.3) chanted Icaros have a completely different overall feeling to the Icaro produced by the Berimbau. Perhaps as an extension of the ‘techniques of ecstasy’ notion (Eliade; 1964), Mariano is changing instrument to change the effect the Icaro is having on the participant, in a way that is similar to Molina’s (2014) notion of ‘changing the boundaries’ of the *realm* being visited by Ayahuasca participants.

4.5 - Structure

There is a lot of symmetrical, palindromic structures, but most examples have required some forced perspectives to observe that pattern, such as in figure XXII. There have been instances where structural content has been difficult to assess due to the limitations of the recordings (3.1, 3.3, 3.5, 3.6). With this in mind, it is clear that to further assess the significance of the structure, much more must be known about the corresponding events that are occurring during the ceremony.



[Image VII: Meditating](#)

Chapter 5

Further Discussion

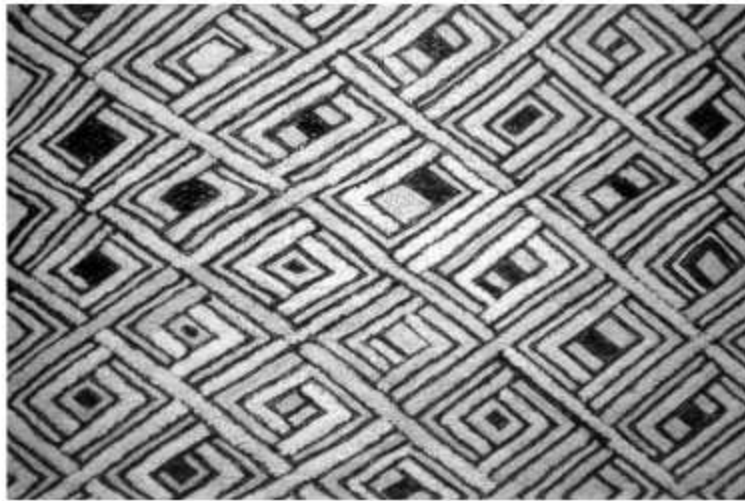
5.1 - Evaluation of the study

In order to evaluate the methodology at this late stage, the research intention must be reconsidered. The original task was to determine a detailed understanding of ‘Ayahuasca Icaro’ musicology, and the success of the methodology can be considered in relation to this. It was achieved, to the extent that general melodic themes of Ayahuasca Icaros have been adequately understood. Contradicting the methodology, handling the melodic content through the use of transcription felt much more comfortable than had been expected. However, limited observations in the rhythmic content suggests that comprehending rhythm requires a new method.

As a suggestion as to which direction the analysis of Ayahuasca Icaros may pursue, it will be interesting to consider the lyrical content in a similar way to Caldicott (n/a). In this Ph.D thesis, the author examines the language used in Icaro chanting. Investigating lyricism in relation to the melodies could reveal particularly powerful words that have been scripted to accentuate particular melody lines.

5.2 - The theme of ‘variation’

Variation stands out as a strong musical characteristic of Ayahuasca Icaros, which exhibit minimal changes on a large scale, creating evolving melodies and rhythms and changing structures. Previously mentioned research (Maas, et al., 2007) also found variation to be a significant characteristic in Gabonian hallucinogenic ritual music. The article uses a picture to demonstrate how subtle variation of simple motifs can create a vast tapestry that is repetitive but evolving and interesting.



[Figure XXX:](#) Image demonstrating the artistic effect of minor variations in a large, repetitive tapestry (Maas et al., 2009)

Maas describes this as ‘periodic repetitions with continuous minimal changes’, and suggests that there is specific reason for utilising this music for the intended purposes. When following a repetitive pattern, the listener begins to generate expectations of each musical idea and, due to the constant minor variations, have them systematically deceived. This circumvention, as suggested by a Mitsogho informant (2007), keeps the listener in a state of ‘open attention’⁵¹. This African tribe uses music to set the listener to the appropriate mental state that is required for the psychological and spiritual healing.

In the same way, Ayahuasca shamans may use minor variations their music to generate musical expectations, and deliberately contradict these expectations; preventing the mind from being able to gain any familiarity, causing it to stay transfixed on the music as the brain searches for patterns and repetitions. It could be suggested that consistent use of familiar frequencies (such as the recurring pitch ‘C’) are used to set up expectations in the mind of the listener, followed by other techniques such as gradual movement in tuning and melodic variation to contradict those expectations.

5.2 - Psychedelic Psychology

Maria Chichosz of the University of Toronto writes about the concept of concentration, and the ‘power’ of paying attention, suggesting that much of the value of ‘tripping’ stems from being forced a serving of (‘ethical’) attentiveness (Chichosz, 2014). Chichosz’ case is such:

⁵¹ Another example of Chichosz’ ‘attentiveness’ a theme discussed in section 4.3

‘Attentiveness’ proves to be an ethical effort, because it requires us to unhinge the natural, hardwired default state of staying alert, instead of becoming ‘hypnotized by the constant monologue in one’s head’⁵². If the Ayahuasca Icaros are causing the effect described in [5.2](#)⁵³, ayahuasca-drinker appears to be shepherded towards this state of ‘ethical attentiveness’ (Chichosz, 2014).

Similar psychological mechanisms are demonstrated in Colombian *Tomas de Yajé*. This ceremony focuses on the consumption of a plant-brew known as *Yagé*⁵⁴. While the ceremony follows a slightly different conduct, the central principles of consuming a *Banisteriopsis Caapi* brew, and being subjected to extensive musical content for the purposes of healing. In simple terms, it is believed that music and sound construct the place which is visited during the ‘trip’. (Molina, 2014) The *taitas*⁵⁵ of Colombian *tomas de Yajé*, use sound tactically to create and modulate temporary, sonically-bound ‘places’ which the mind of participants travel to during the ceremony. Molina writes of ‘placeness’, and how non-verbal sound is used to index the sounds of the Amazon (Rivers, Wind, Animals), to anchor the participants in their location. This could stand as an example of Maria Chichosz’ (2014) ethical attentiveness, using sonic tactics to help define the participants sense of ‘place’ during their incorporeal travels. Molina notes a valuable area of further study, in understanding the nature of which non-verbal sound defines ‘place’, somewhat combining the realms of semantic (such that it contains information useful to the observer) sonic information and musical sonic information to greater understand just how much external frequencies interact with internal anatomical or cerebral frequencies.

5.3 - Jungle Gym/Spiritual Ladder

The idea of ‘anchoring’ discussed in [5.3](#) ties directly into the aforementioned notion of spiritual ladders and ropes (Maas, et al., 2009; Beattie, J., Middleton, J., 2004; Narby, 1999; Eliade, 1964). It has been previously proposed that the music associated with hallucinogenic ritual acts as a guide rope between one realm and another (perhaps one that has been ‘created’⁵⁶),

⁵² The internal monologue refers to the conversation that the mind appears to undergo when evaluating and assimilating the direct information of the present moment, Chichosz suggests that our contextual understanding of things form a backdrop to the information being perceived in the present moment, a mechanism which prevents us from the aforementioned hypnotization or ‘entrancing’.

⁵³ Whereby the music prevents the brain from finding a true rhythm, causing it to stay search

⁵⁴ This plant-brew shares chemical composition with traditional Amazonian Ayahuasca, such that it is comprised of *Banisteriopsis Caapi* and *Psychotria Viridis* (or the many variations). The conduct of the ceremony is different, as they likely have a different cosmological perspective and thus engage with the spirits in a unique manner. However different though, fundamental principles of Plant Spirit communication and dependence on music remain ever present in *Tomas de Yajé*

⁵⁵ *Taitas* is the Colombian term for the musical shaman that conducts *Tomas de Yajé*.

⁵⁶ Referring back to [5.2](#), where it was suggested that the *taitas* in Colombian *Tomas de Yajé* use music to create a ‘sonically bound’ dimension.

such that it anchors them in reality, assisting a safe return and acting as a permanent link back to a state of normal waking consciousness. Once again referring to [5.2](#), Molina (2014) suggests that Colombian *taitas* use their sonorous instruments to create temporal ‘places’ that are ‘sonically bound’, and modulate their boundaries while the participants explore them. This shares a striking familiarity to the *Jungle Gym in Consciousness* concept developed by Katz and Dobkin de Rios, where they propose that the shamanic Icaros sung during Ayahuasca ceremonies act as a self-contained structure for the participants. While the findings of this research aren’t enough to further shape this theory, 35 years after it’s conception, modern Ayahuasca related research continues to confirm the idea that Shamans create a plane in which the user’s ecstatic consciousness can exist within.

5.4 - Final Words

The final area of discussion includes the potential this research has to go beyond its current field of study. Research by Carhartt-Harris (2014) suggests that psychoactive substances can interfere or modulate with the default mode network (DMN)⁵⁷. The concept of the ‘ego’ is currently believed to be an emergent property of the DMN, and it is considered as the location of metacognition, area of the brain; where the brain ‘thinks about thinking’. If the ego is the internal, hypnotic conversation that accompanies the reality being experienced by everyone, it can be quite clear to envision how interference with the DMN (often described as ego-dissolution), can drastically affect how reality is perceived⁵⁸. The neurobiological understandings of consciousness are beyond the scope of this paper, however, the results found in this musicological research can be useful to researchers who study the synapse between music and its manifestation as a change in consciousness⁵⁹. Section [5.2](#) demonstrates one direction of further study that the findings of Ayahuasca Icaros could pursue, and the further neuroscientific discussions emerging around these psychological questions demonstrate that these results directly feed into other fascinating research.

⁵⁷ This DMN has been dubbed ‘the orchestrator of the self’ (Carhartt-Harris, 2014)

⁵⁸ Due to the loss of the inner-commentary, or, as Chichosz (2014) suggests, a force-serving of ethical attentiveness

⁵⁹ Carhartt-Harris (2014) further states, in a short online lecture, that all positive and negative experiences had during the psychedelic experience can have therapeutic and psychological benefits should it be properly mediated (Sessa, 2008). This makes clear the importance of the shaman role within ayahuasca ceremonies, and highlights the importance of further studying ayahuasca icaros, as they are the main tool with which the shaman mediates the experience.

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6.2 - Images

Image I: Shaman at work -

http://paranormal-news.ru/news/kak_shamany_peru_otkryvajut_soznanie_i_isceljajut/2014-10-01-9832

Image II: The River Amazon - <http://www.thousandwonders.net/Amazon+River>

Image III: Brewing Ayahuasca -

https://www.erowid.org/chemicals/show_image.php?i=ayahuasca/ayahuasca_brew_pot_i2014e0083_disp.jpg

Image IV: Preparing Ayahuasca Brew -

<http://botanicsscience.blogspot.co.uk/2011/01/etnofarmakologia-i-dopalacze.html>

Image V: Ready to Brew - <http://www.justsciencenews.com/2015/04/06/page/3/>

Image VI: Fanning the flames -

<http://www.mensjournal.com/magazine/down-the-monkey-hole-20140310?page=4>

Image VII: Meditating - <http://www.incaglossary.org/s.html>

6.3 - Recordings Used

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Note: There is worked out to be 2,500 words in figure titles, references, contents page, titles, image captions and other. This brings the main body word count to roughly 11,500.